

{3} Current Electricity

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- * INTRODUCTION
- * ELECTRIC CURRENT
- * ELECTRIC CURRENTS IN CONDUCTORS
- * OHM' S LAW
- * DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY
- * LIMITATIONS OF OHM 'S LAW

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* OHM'S LAW

* DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY

* LIMITATIONS OF OHM'S LAW

* TEMPERATURE DEPENDENCE OF RESISTIVITY

* ELECTRICAL ENERGY, POWER

* CELLS, EMF, INTERNAL RESISTANCE

* CELLS IN SERIES AND IN PARALLEL

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- * **Charges in motion constitute an electric current.**
- * In the present chapter, we shall study some of the basic laws concerning **steady electric currents**.

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- * The direction of current is taken as the direction of motion +ve charges or opposite to the motion of -ve charge.

Ex 1 : In cosmic rays 0.15 protons/cm²-s are entering the earth's atmosphere. If the radius of the earth is 6400 km, what is the current received by the earth in the form of cosmic rays?

Ex 2: In neon gas discharge tube 2.9×10^{18} Ne^+ ions move to the right through a cross-section of the tube each second, while 1.2×10^{18} electrons move to the left in this time. What is the net electric current in the tube?

Ex 3 : In the Bohr's model of hydrogen atom, the electron is assumed to rotate in a circular orbit of radius 5×10^{-11} m, at a speed of 2.2×10^6 m/s. What is the current associated with electron motion?

Ex 4 : The current in a wire varies with time according to the relation $I = 2 \text{ A} + (0.6 \text{ A/s}^2) t^2$.

- (a) How many coulomb of charge pass a cross-section of the wire in the time interval between $t = 0$ and $t = 10 \text{ s}$?
- (b) What constant current would transport the same charge in same charge in the same time interval?

ELECTRIC CURRENTS IN CONDUCTORS

DRIFT VELOCITY

Electrons in a conductor are in constant random motion.

When an electric field is applied, they acquire a net drift velocity.

The drift velocity is the average velocity of the electrons in the direction of the electric field.

It is denoted by v_d .

The drift velocity is directly proportional to the electric field.

Mathematically, $v_d \propto E$.

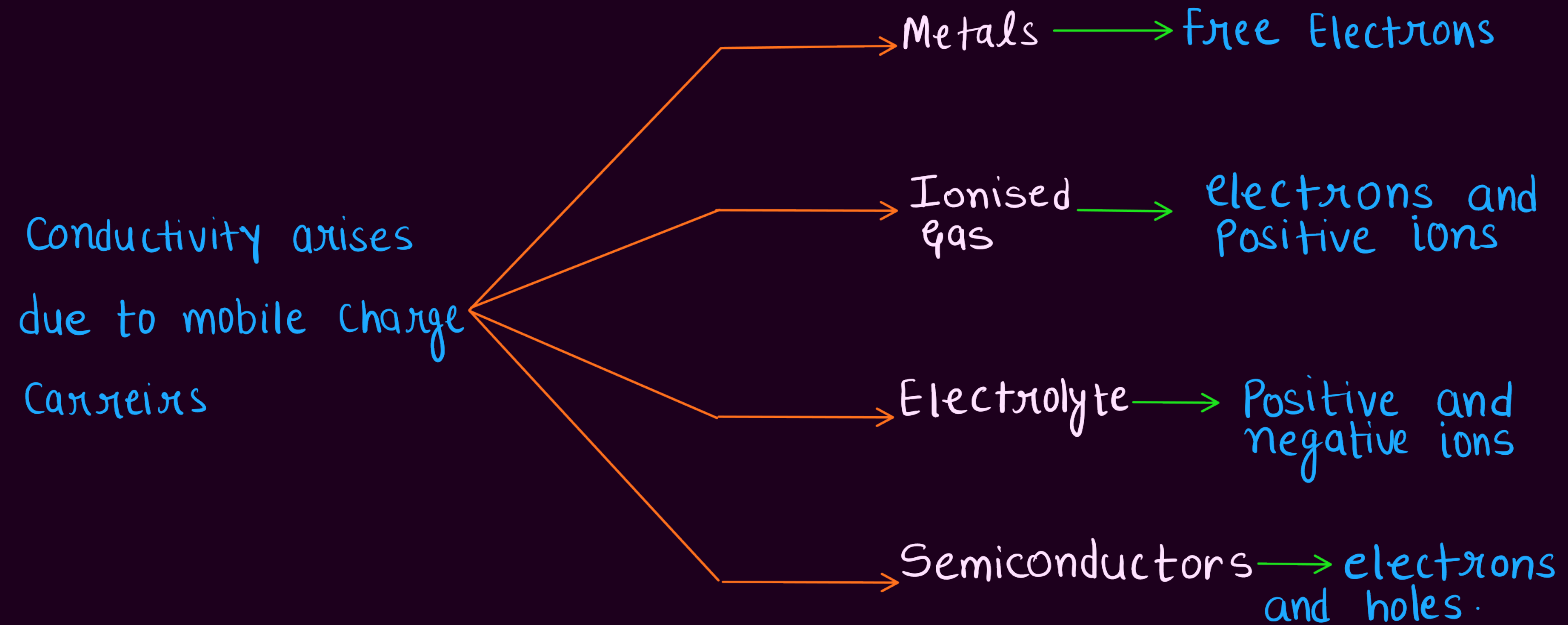
Where E is the electric field.

The drift velocity is inversely proportional to the cross-sectional area of the conductor.

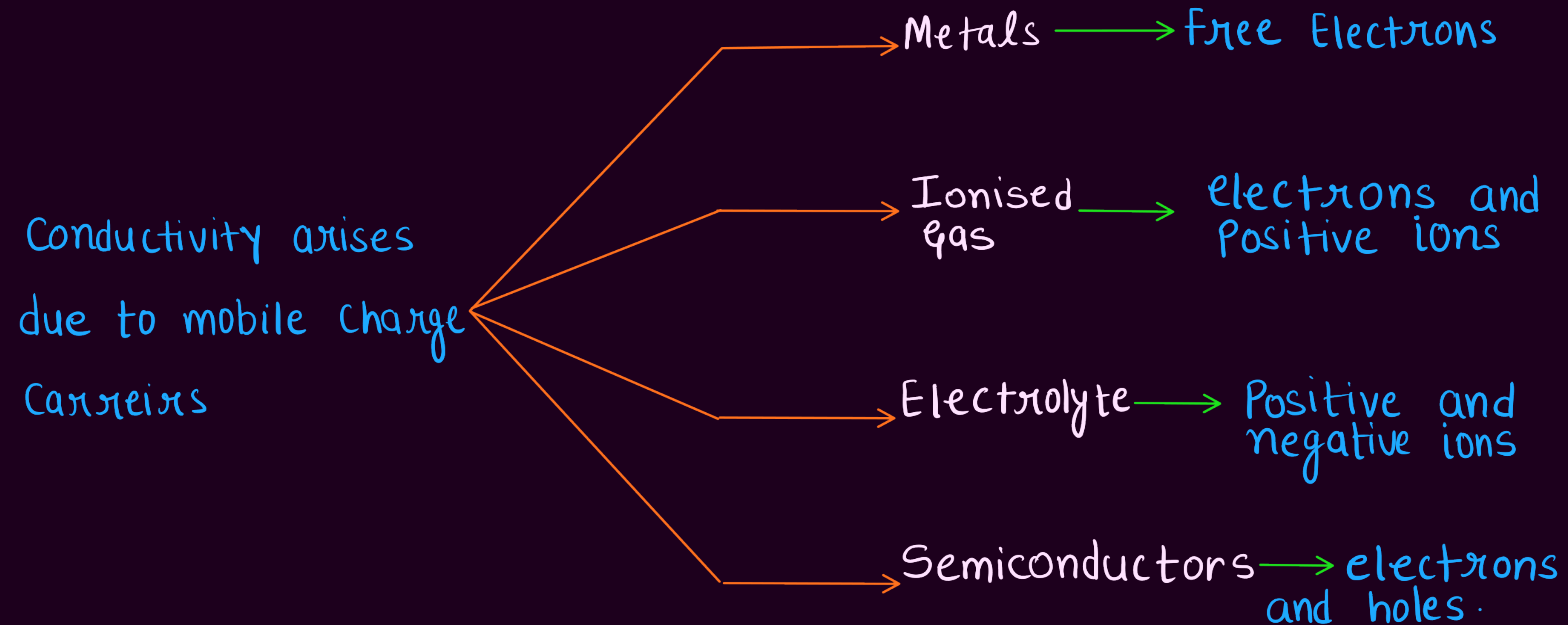
Mathematically, $v_d \propto \frac{1}{A}$.

Where A is the cross-sectional area of the conductor.

ELECTRIC CURRENTS IN CONDUCTORS



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* In our discussions, we will focus only on solid conductors so that the current is carried by the negatively charged electrons in the background of fixed positive ions.

ELECTRIC CURRENTS IN CONDUCTORS

CONDUCTORS AND RESISTANCE

Conductors are materials that allow electric current to flow easily. They have a low resistance to the flow of electric charge.

The resistance of a conductor is a measure of how much it opposes the flow of electric current. It is denoted by the symbol R .

The resistance of a conductor depends on its material, length, and cross-sectional area. It is given by the formula:

$$R = \frac{\rho L}{A}$$

where ρ is the resistivity of the material, L is the length of the conductor, and A is the cross-sectional area.

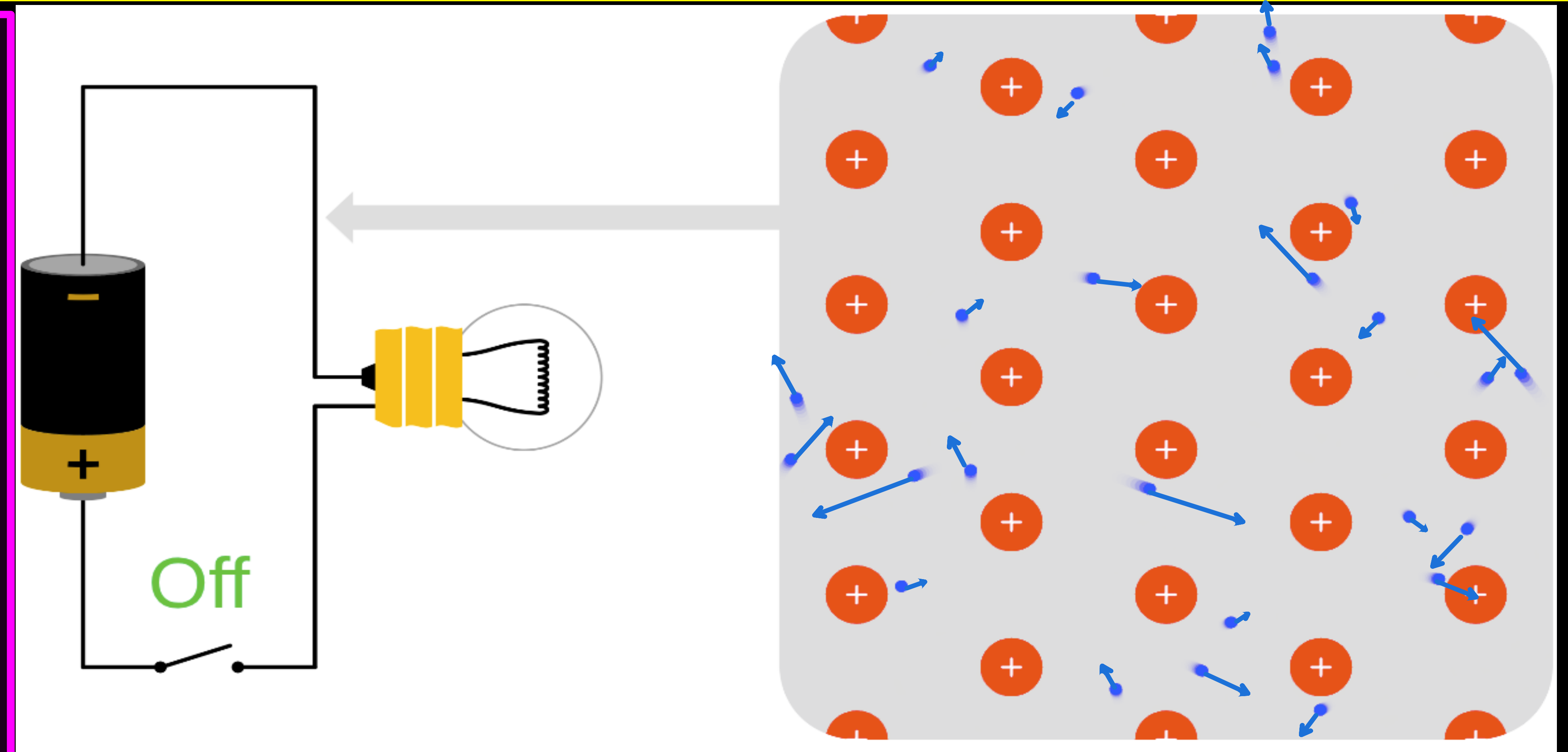
The resistance of a conductor is directly proportional to its length and inversely proportional to its cross-sectional area.

The resistance of a conductor is also affected by temperature. For most materials, the resistance increases as the temperature increases.

The resistance of a conductor is a property that is independent of the voltage applied across it or the current flowing through it.

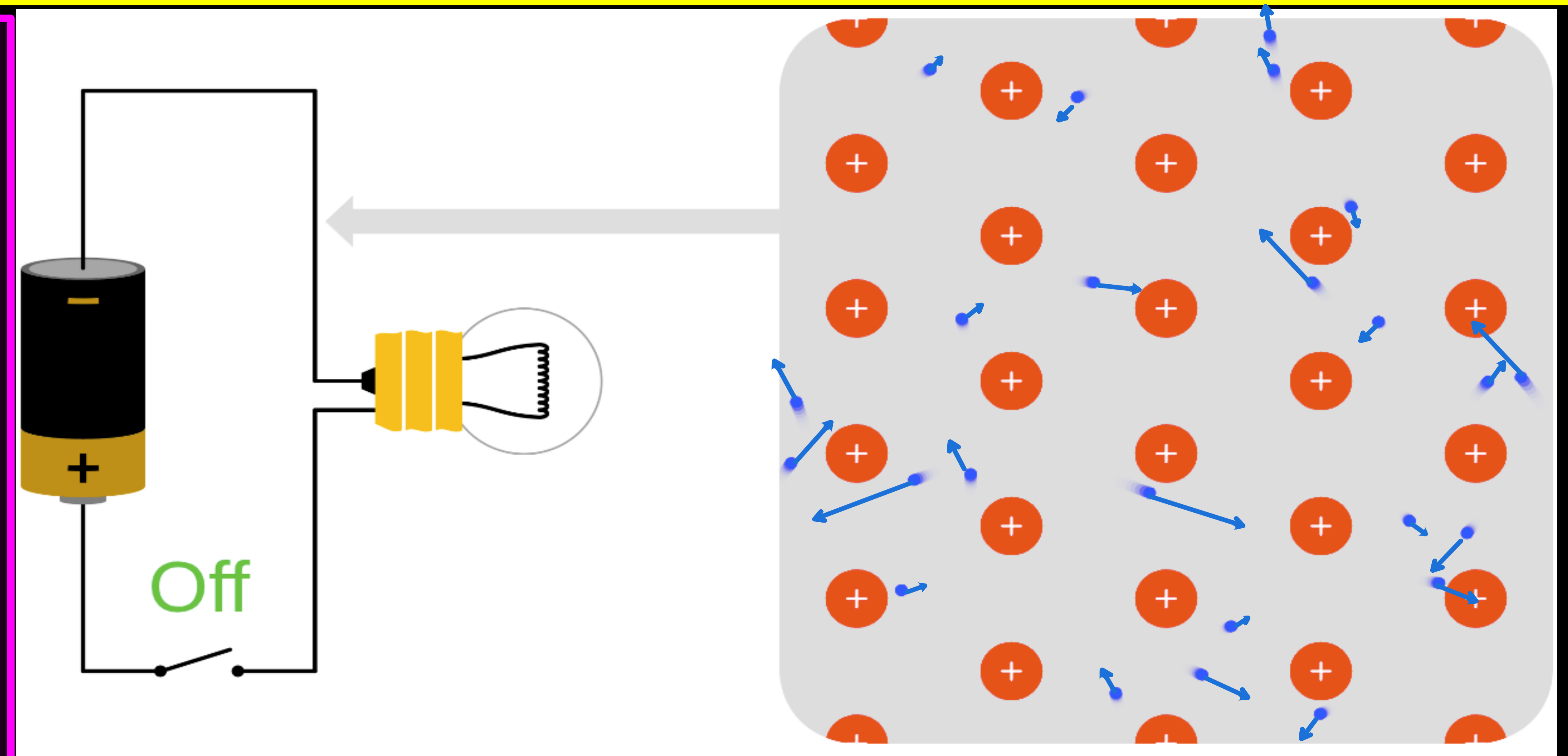
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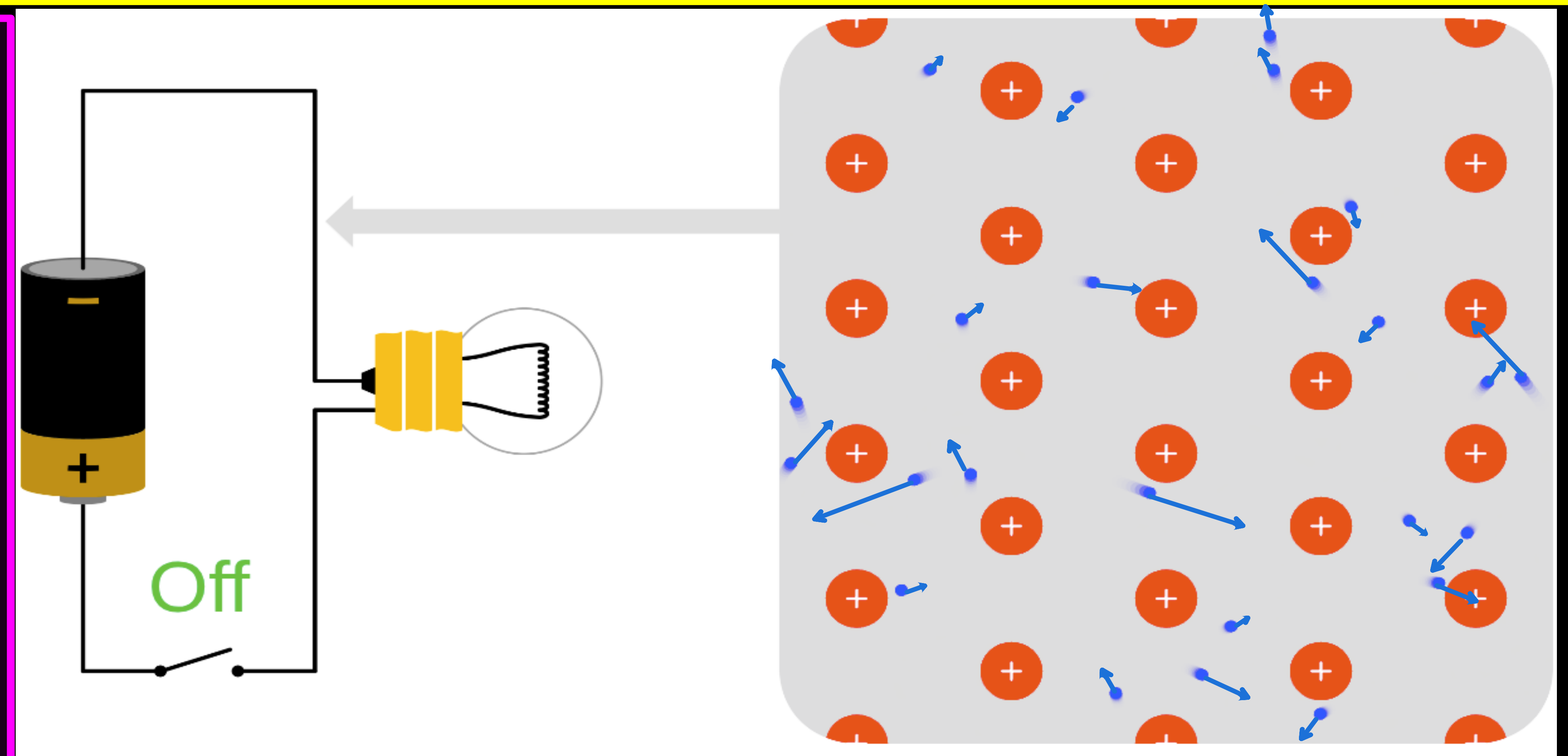
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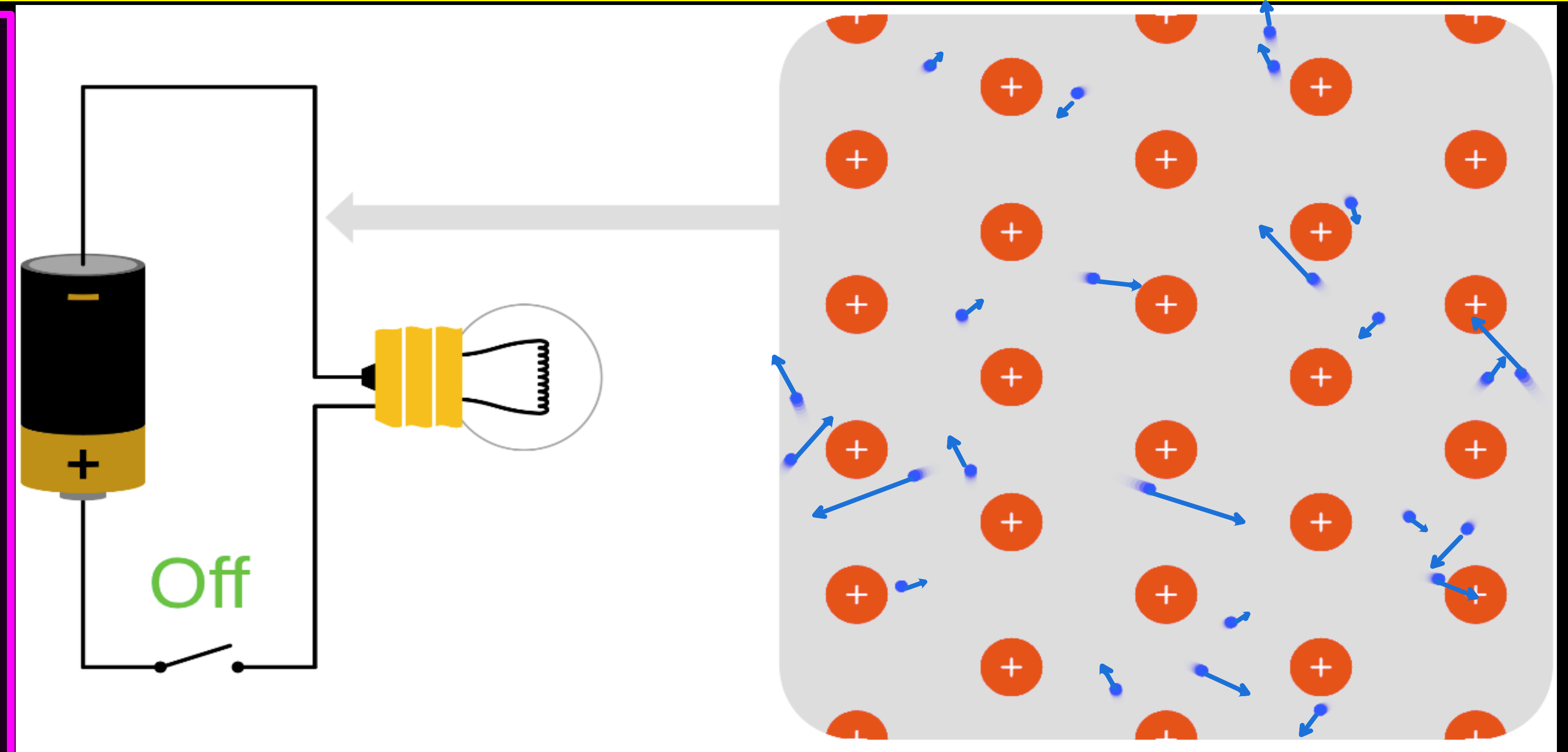


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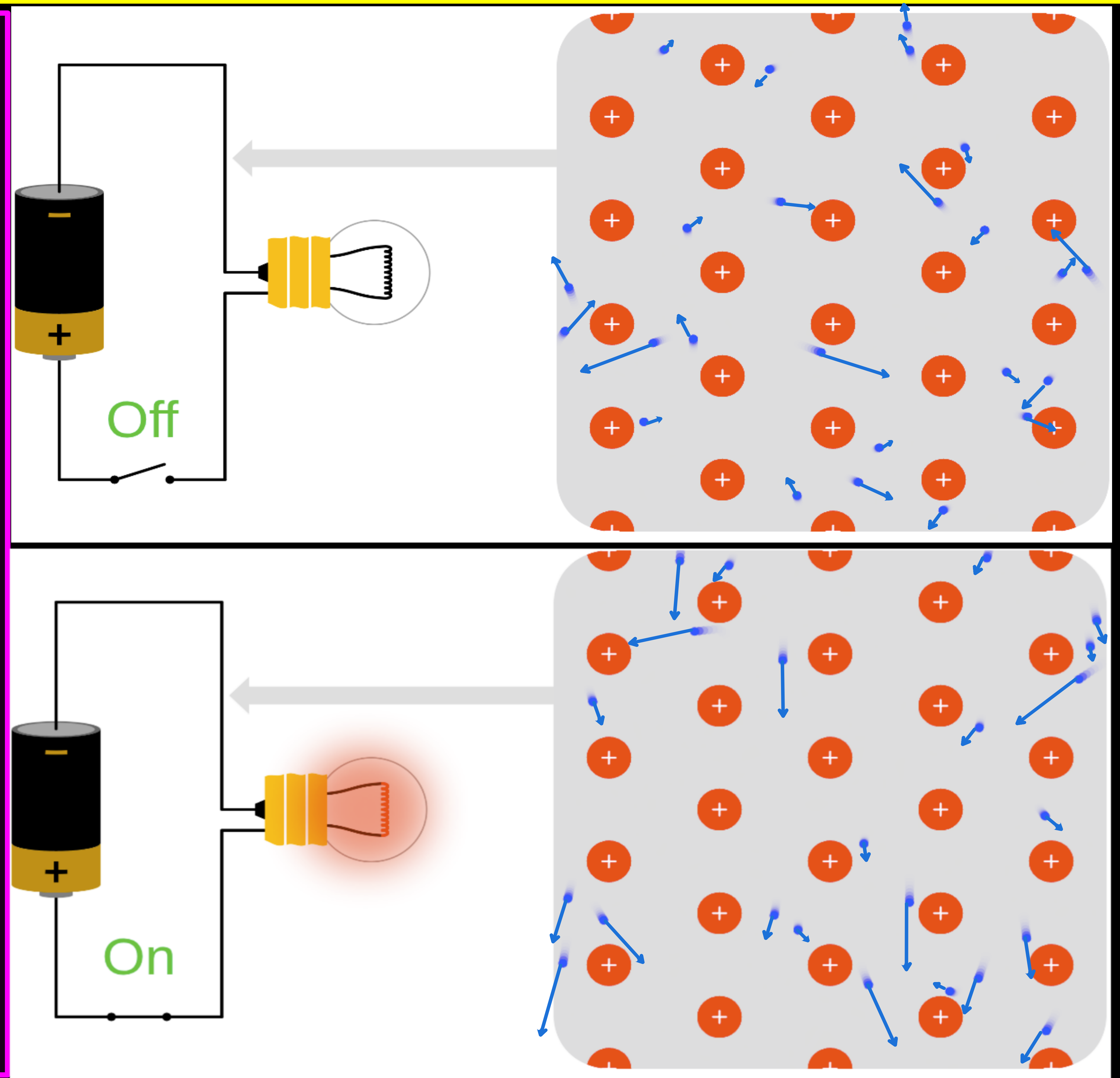
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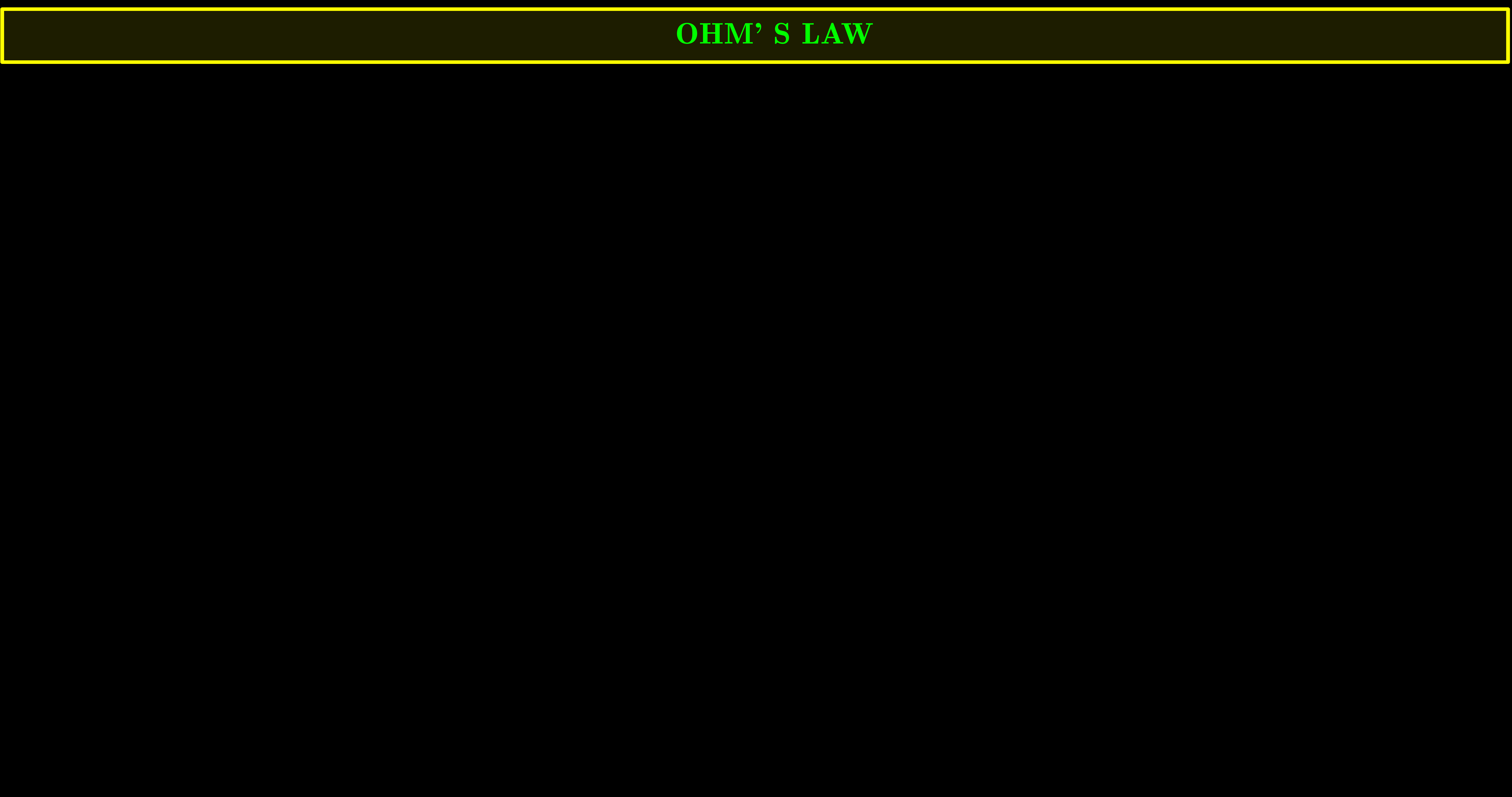
Thus on the average, the number of electrons travelling in any direction will be equal to the number of electrons travelling in the opposite direction. So, there will be no net electric current.

Case 2 : When electric field is present :

* The electrons will be accelerated due to this field, in the opposite direction of the field. The electrons, as long as they are moving, will constitute an electric current.



OHM'S LAW



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Imagine a conductor through which a current I is flowing and let V be the potential difference between the ends of the conductor. Then Ohm's law states that the current in a conductor is directly proportional to the voltage applied:

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$$R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A}$$

* where the constant of proportionality called resistivity (ρ) depends on the material

CURRENT DENSITY AND EQUIVALENT FORM OF OHM'S LAW

Consider a rectangular block of material of length l , cross-sectional area A , and resistivity ρ .

When a potential difference V is applied across the ends of the block, a current I flows through it.

The current density J is defined as the current per unit cross-sectional area:

$$J = \frac{I}{A}$$

The resistance R of the block is given by the ratio of the potential difference to the current:

$$R = \frac{V}{I}$$

Using the definition of resistance and the expression for current density, we can write:

$$R = \frac{V}{I} = \frac{V}{JA}$$

For a uniform material, the resistance can also be expressed in terms of the resistivity ρ , length l , and cross-sectional area A :

$$R = \rho \frac{l}{A}$$

Combining the two expressions for resistance, we get the equivalent form of Ohm's Law in terms of current density:

$$E = \rho J$$

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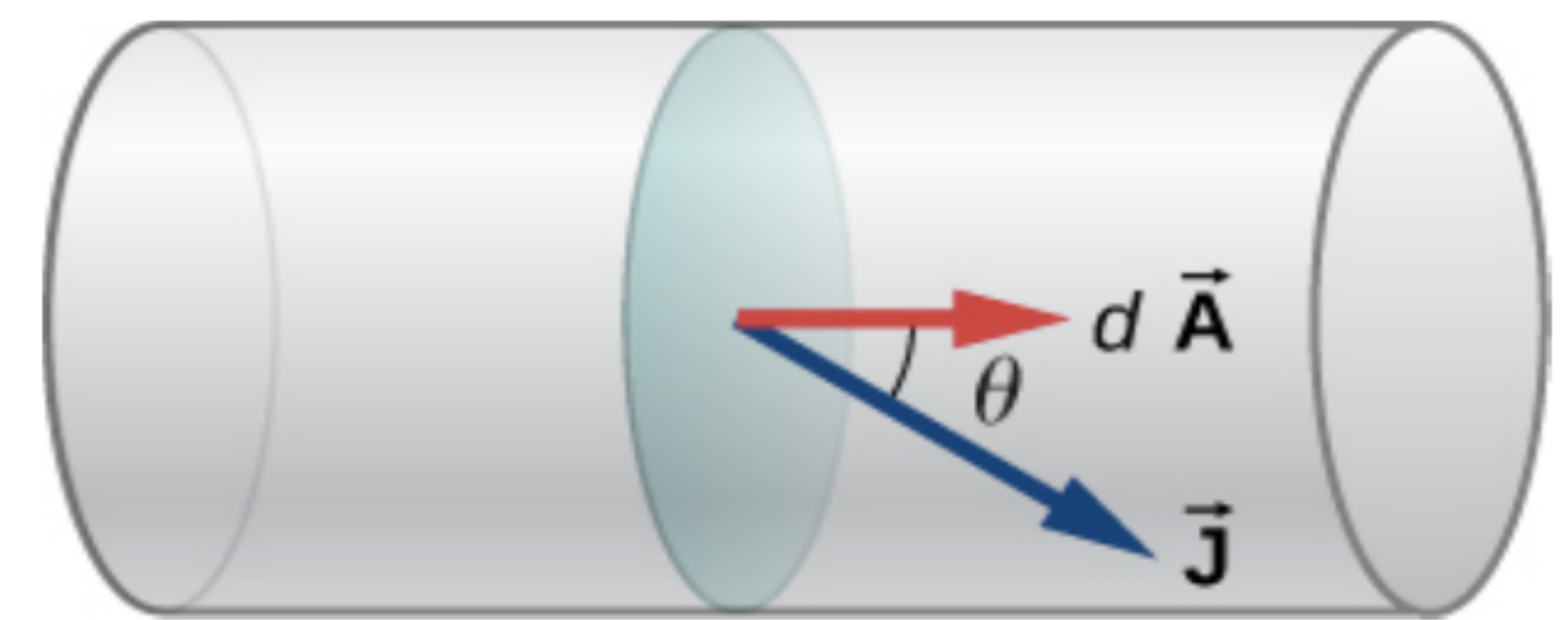


Figure 9.12 The current density $\vec{J}$ is defined as the current passing through an infinitesimal cross-sectional area divided by the area. The direction of the current density is the direction of the net flow of positive charges and the magnitude is equal to the current divided by the infinitesimal area.

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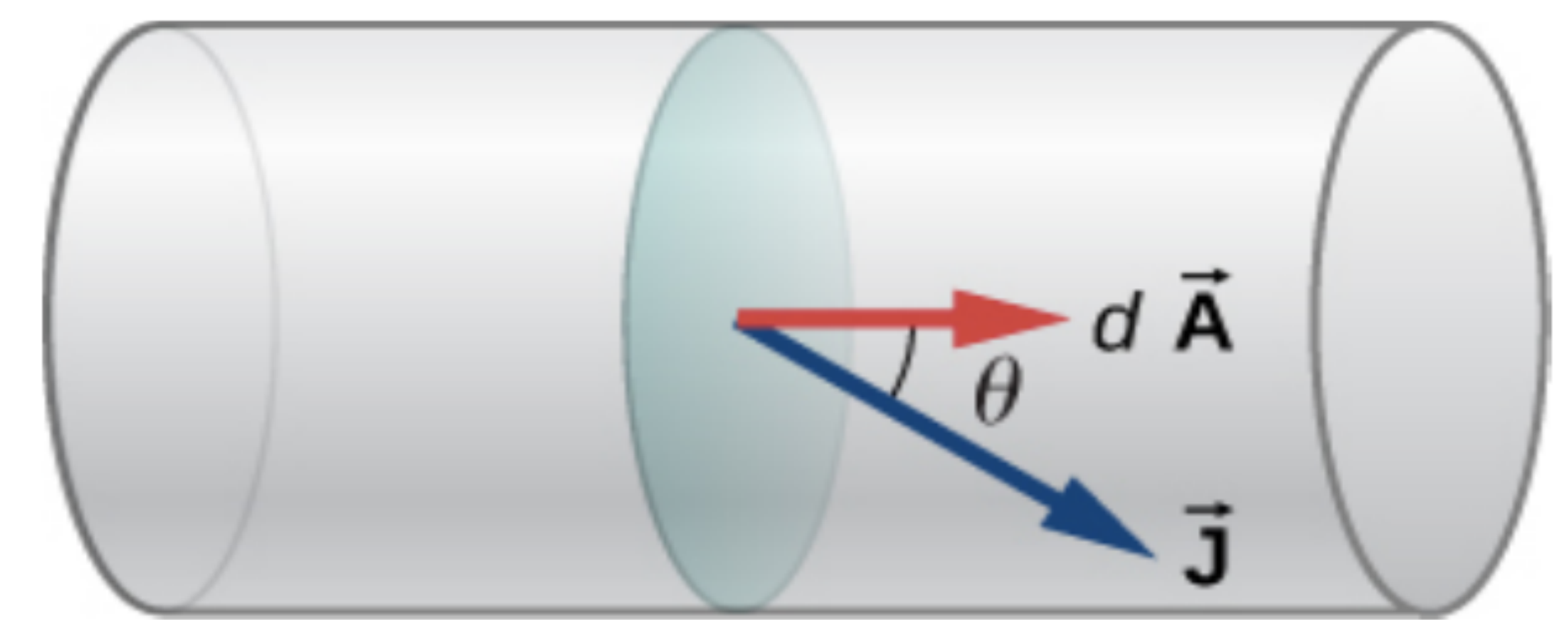


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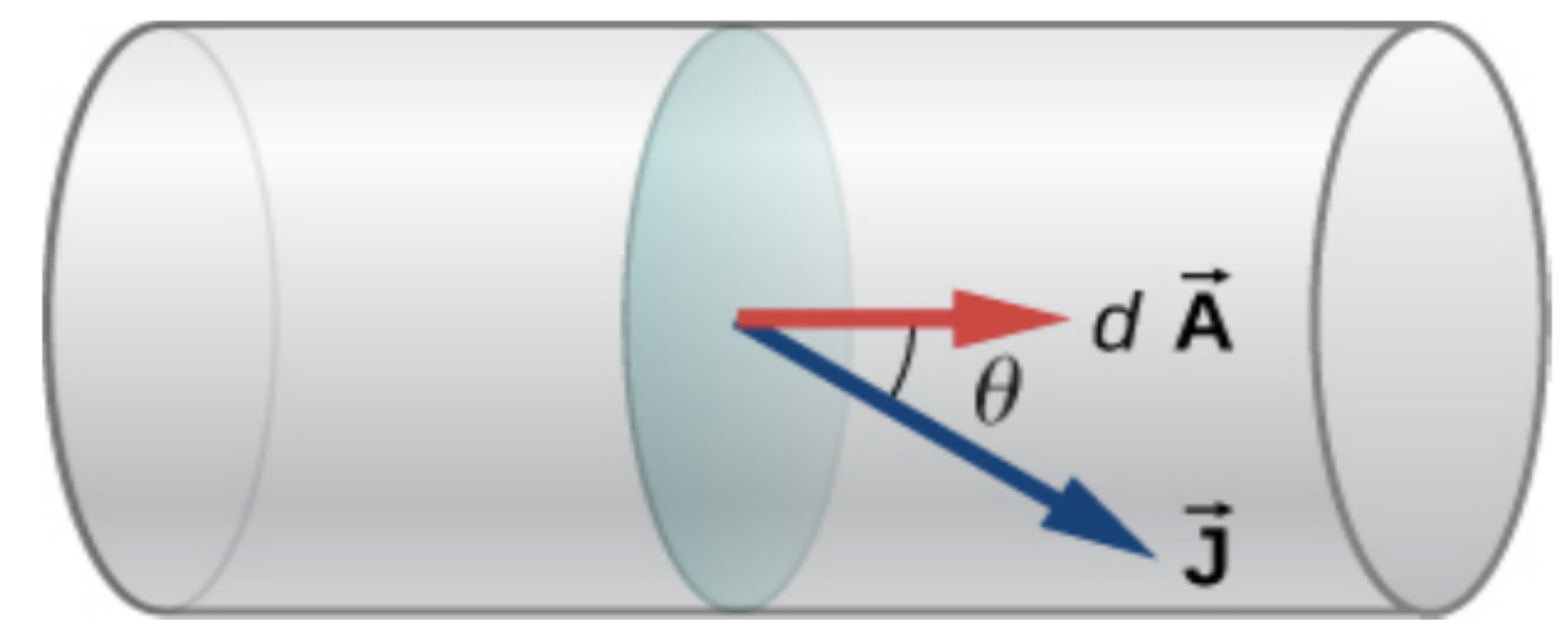


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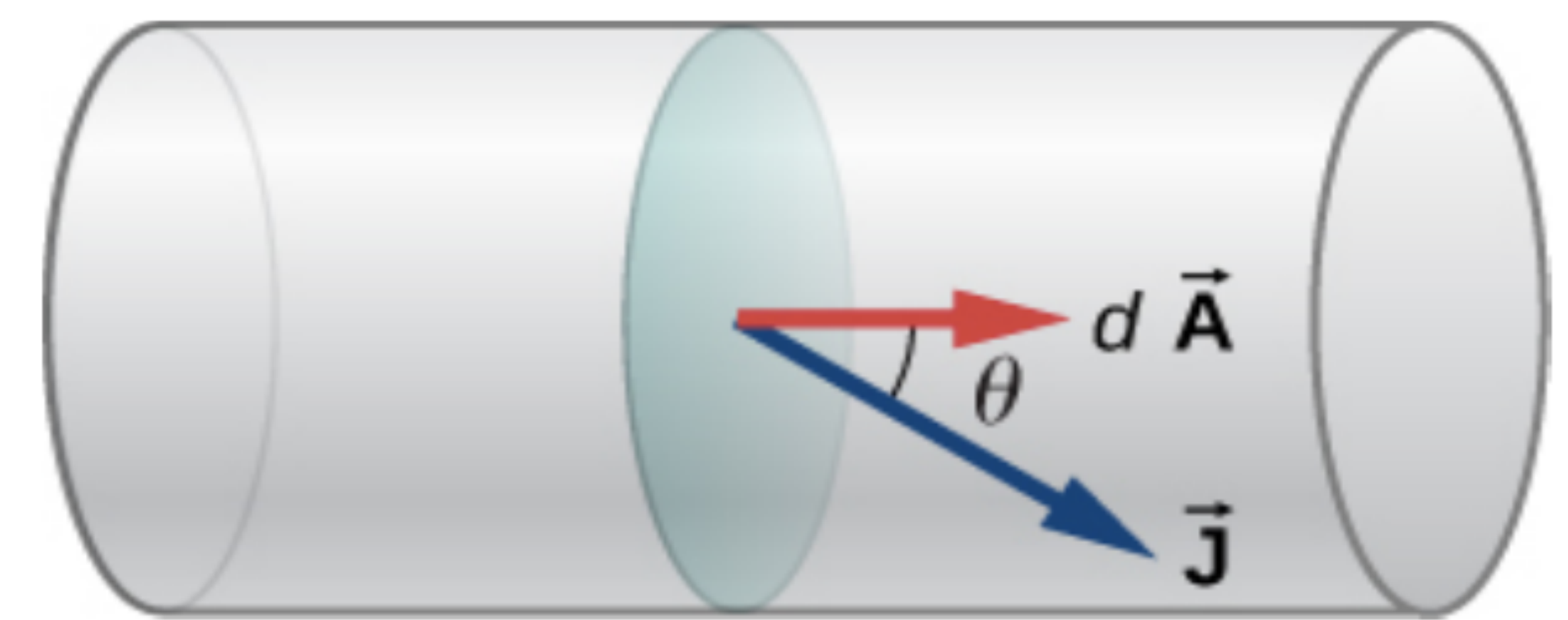


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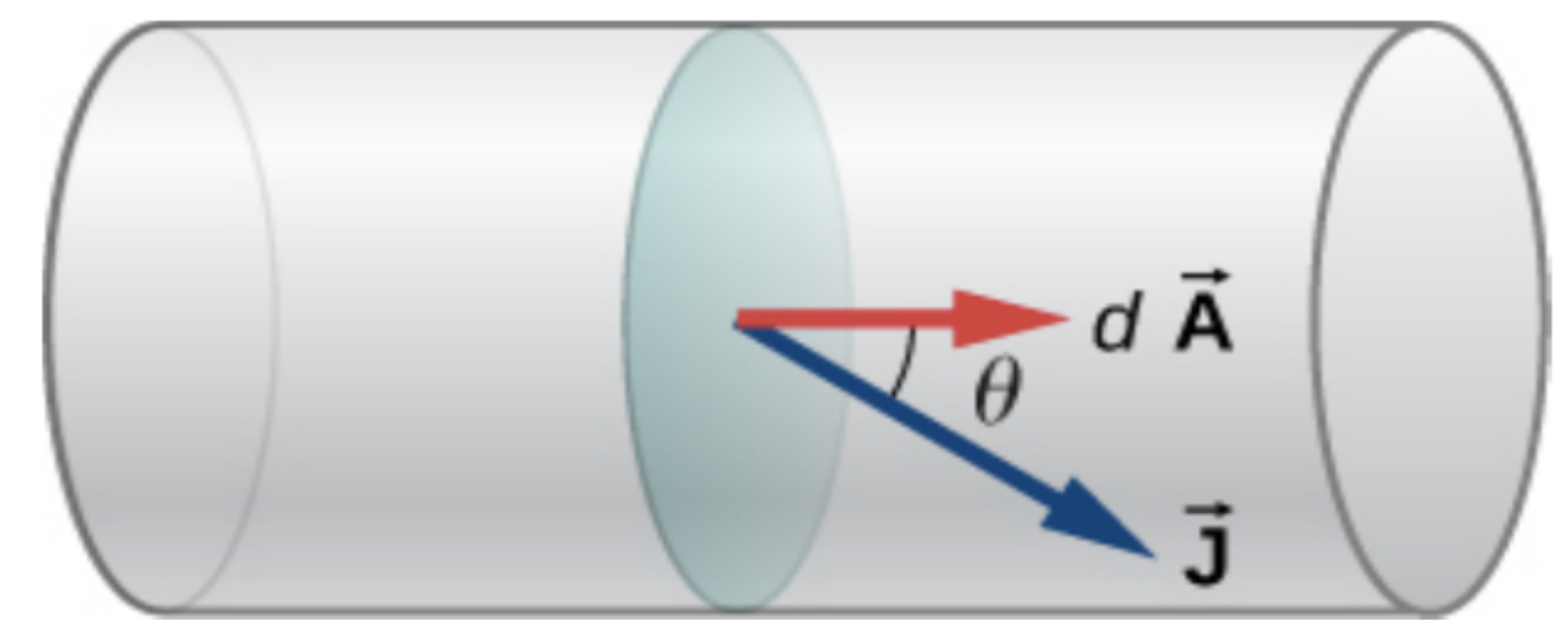


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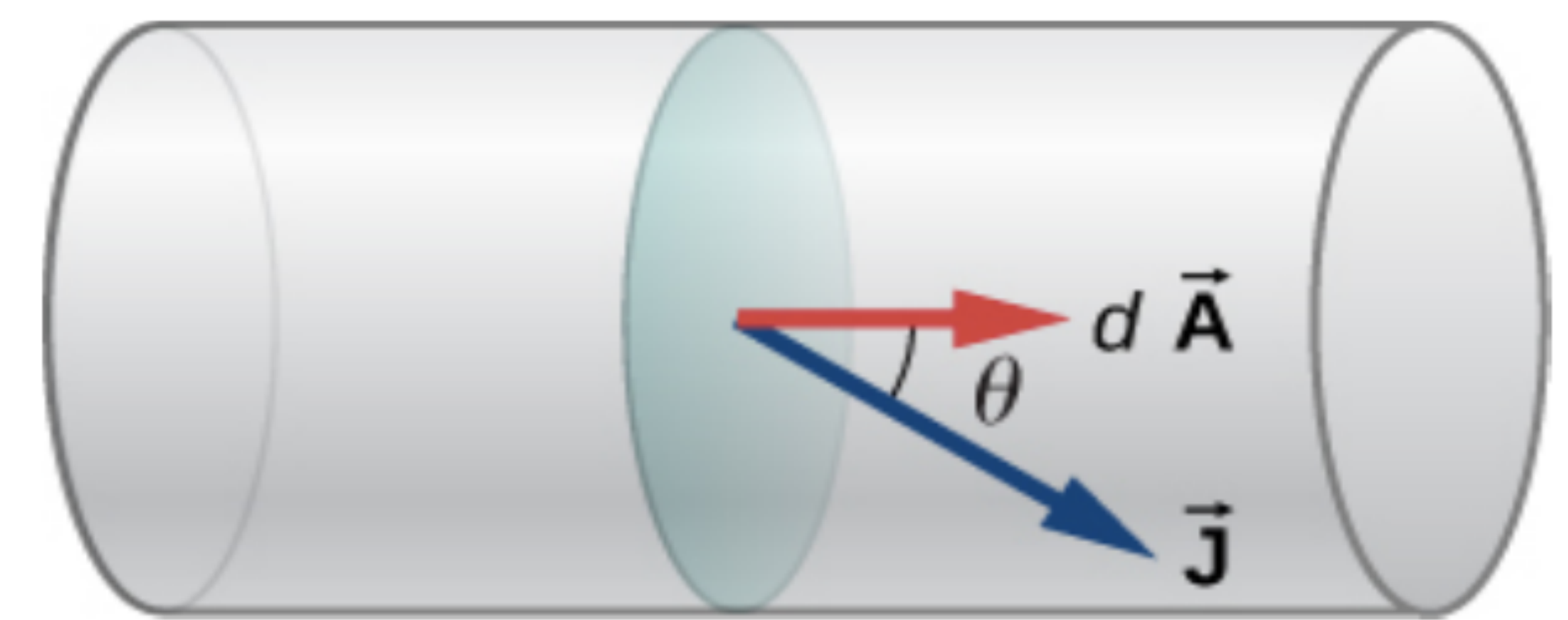


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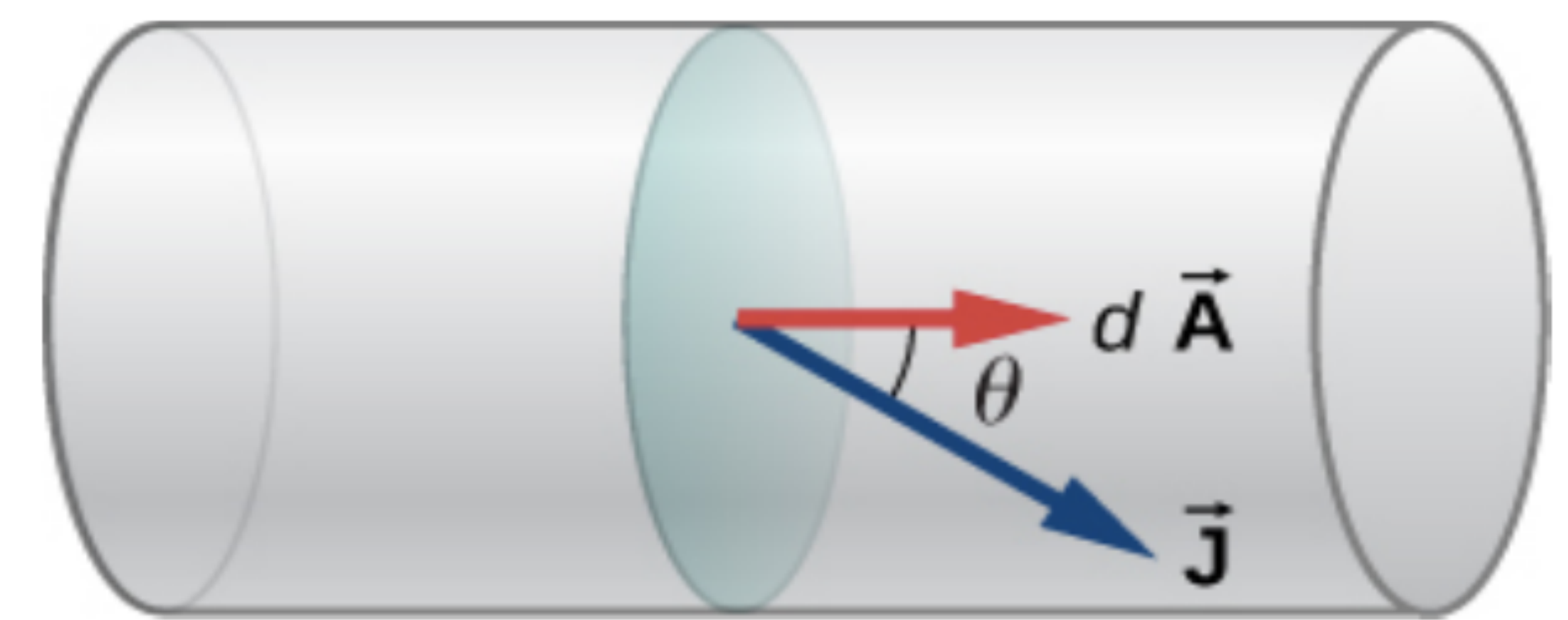


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3.6 A negligibly small current is passed through a wire of length 15 m and uniform cross-section $6.0 \times 10^{-7} \text{ m}^2$, and its resistance is measured to be $5.0 \text{ } \Omega$. What is the resistivity of the material at the temperature of the experiment?

Ex 6 : A wire 50 cm long and 1 mm² in cross-section carries a current of 4 A when connected to a 2 V battery. The resistivity of the wire is

a) $2 \times 10^{-7} \Omega m$ *b)* $5 \times 10^{-7} \Omega m$

c) $4 \times 10^{-6} \Omega m$ *d)* $1 \times 10^{-6} \Omega m$

Ex 7: A wire of length L and cross-section area A has resistance R . What will be the resistance of the wire if it is stretched to twice its original length? Assume that the density and resistivity of material do not change when the wire is stretched.

Ex 8 : A wire of length L is drawn such that its diameter is reduced to half of its original diameter. If the initial resistance of the wire is $10\ \Omega$, its new resistance would be

a) $40\ \Omega$

(b) $80\ \Omega$

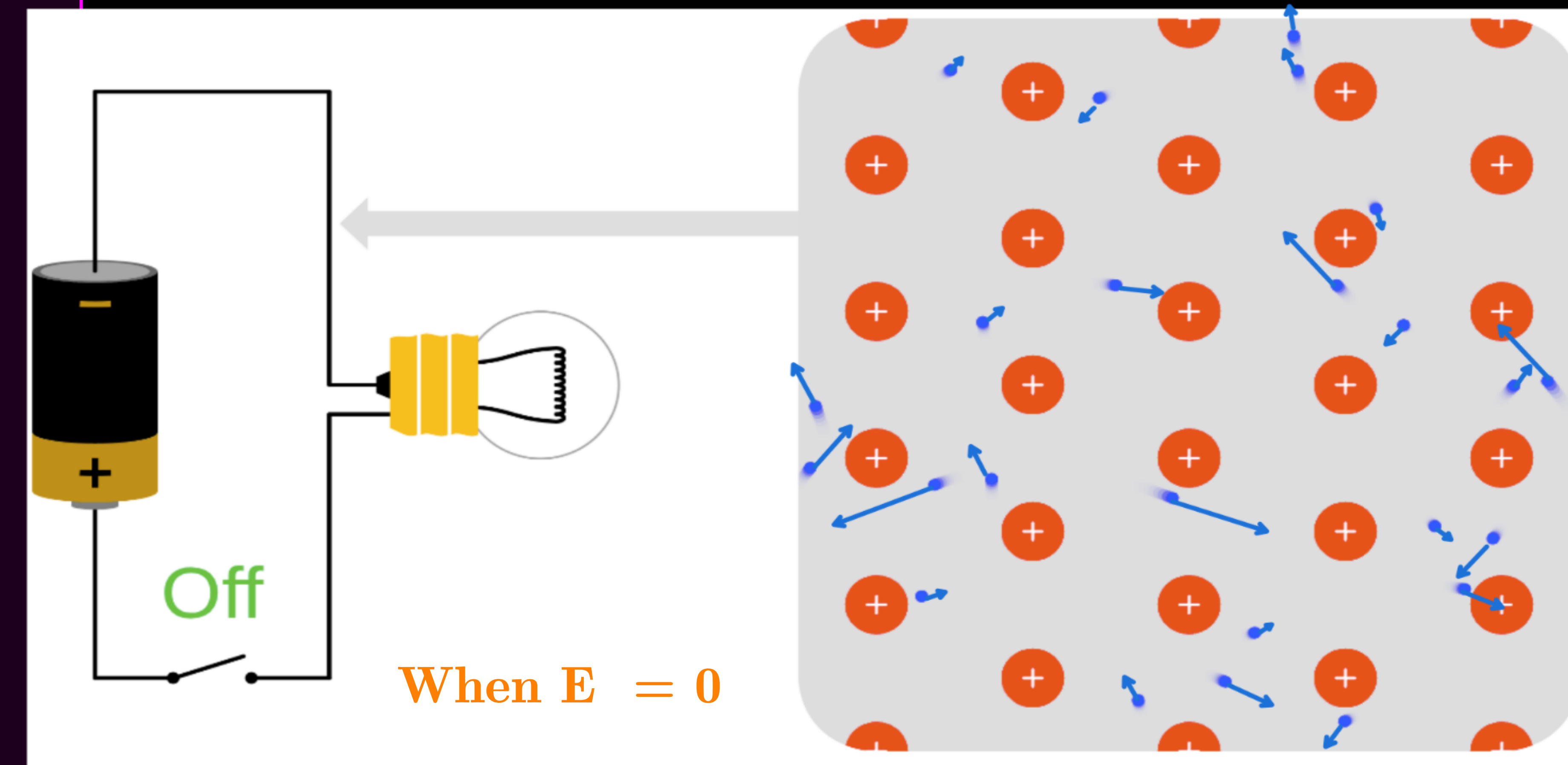
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(d) $160\ \Omega$

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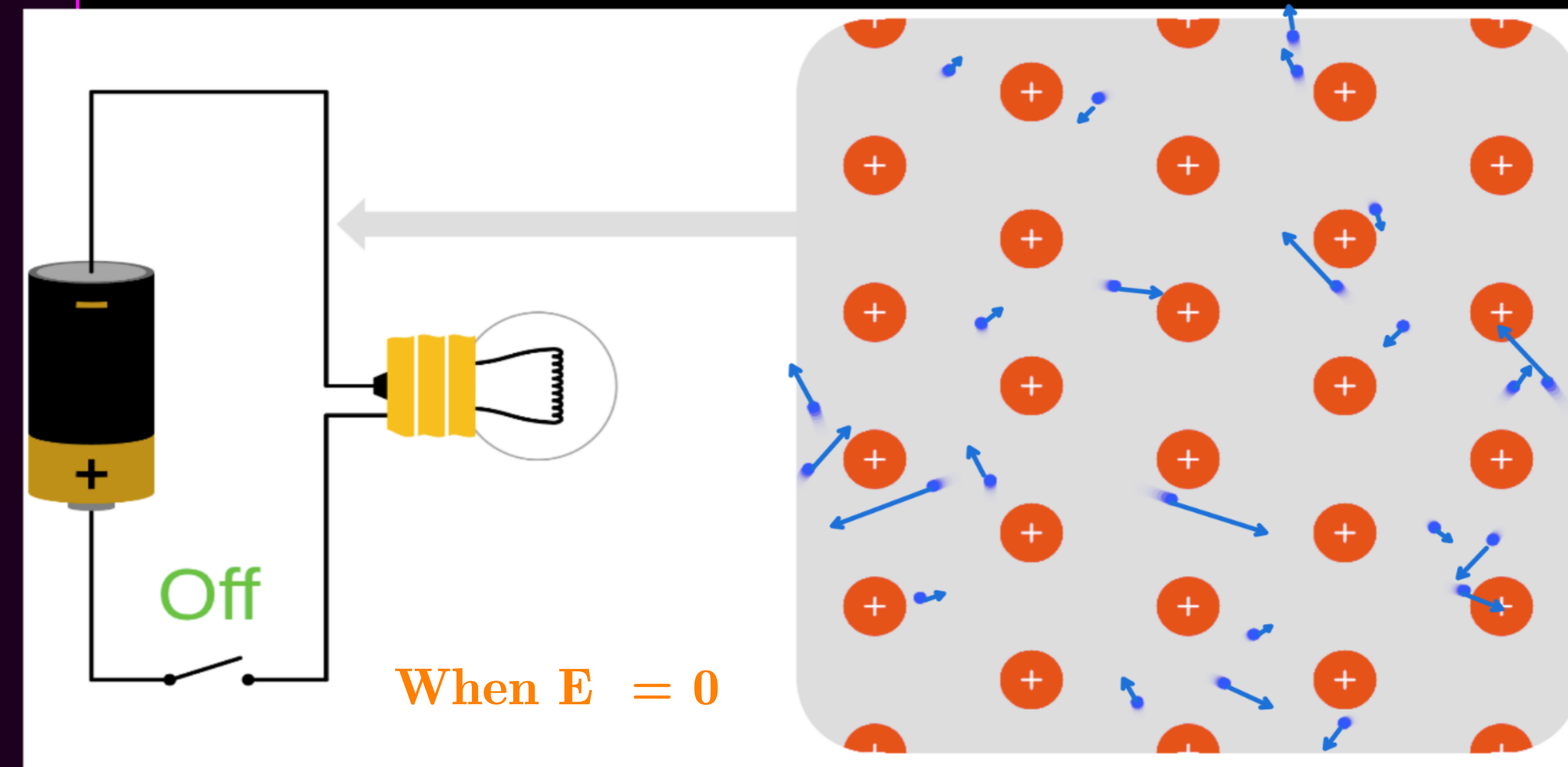
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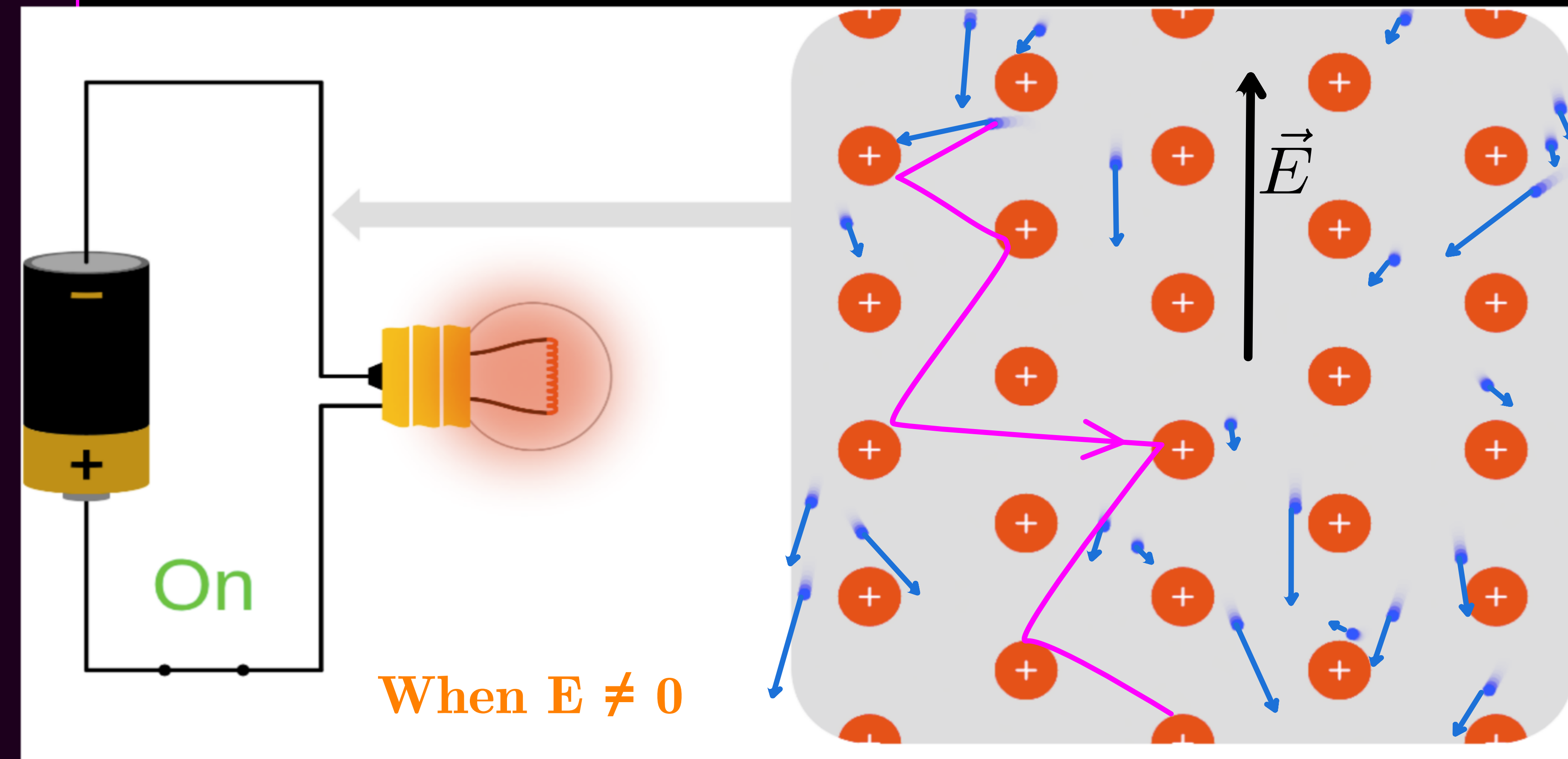
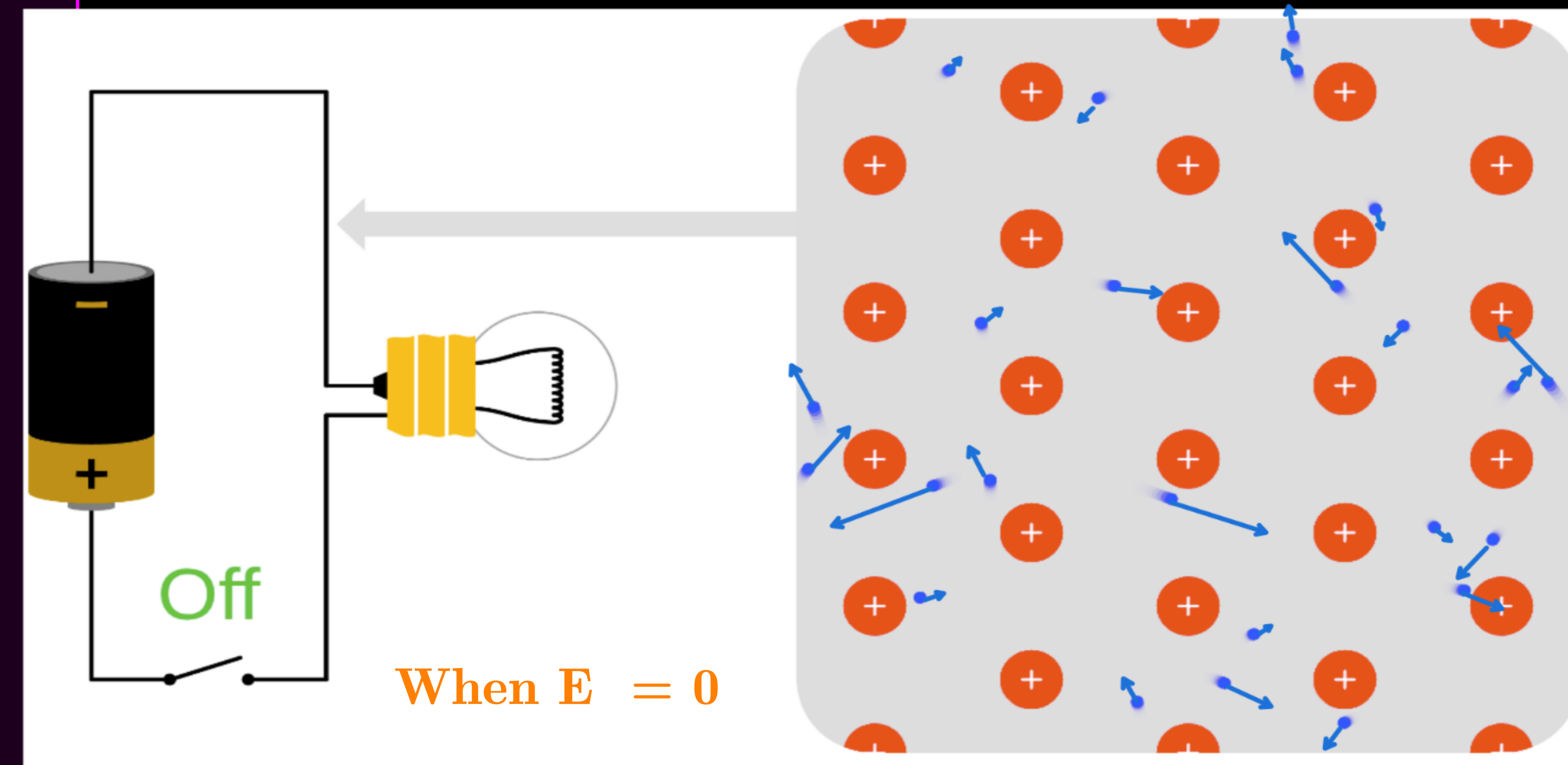
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$$\vec{F} = -e\vec{E}$$



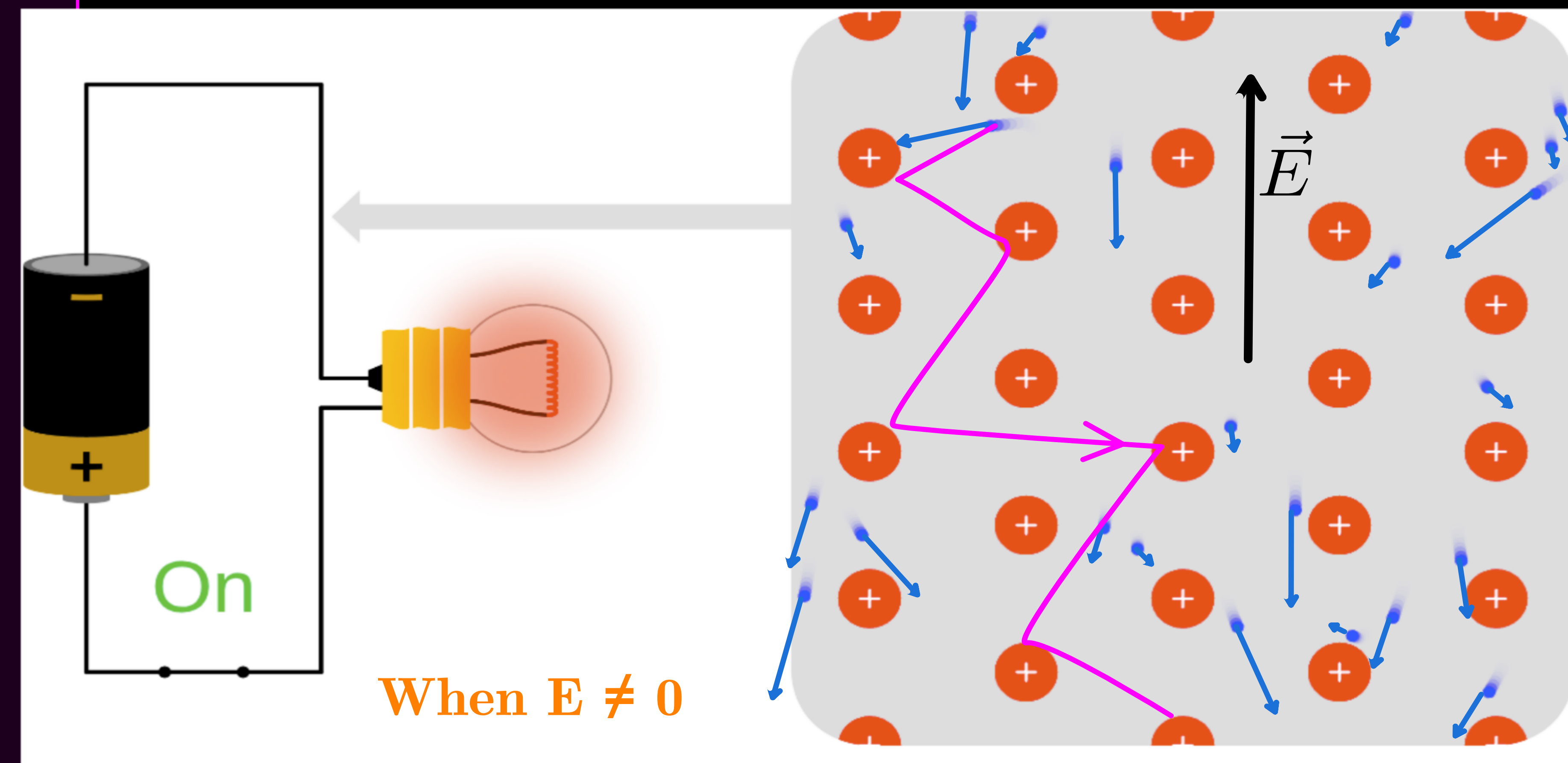
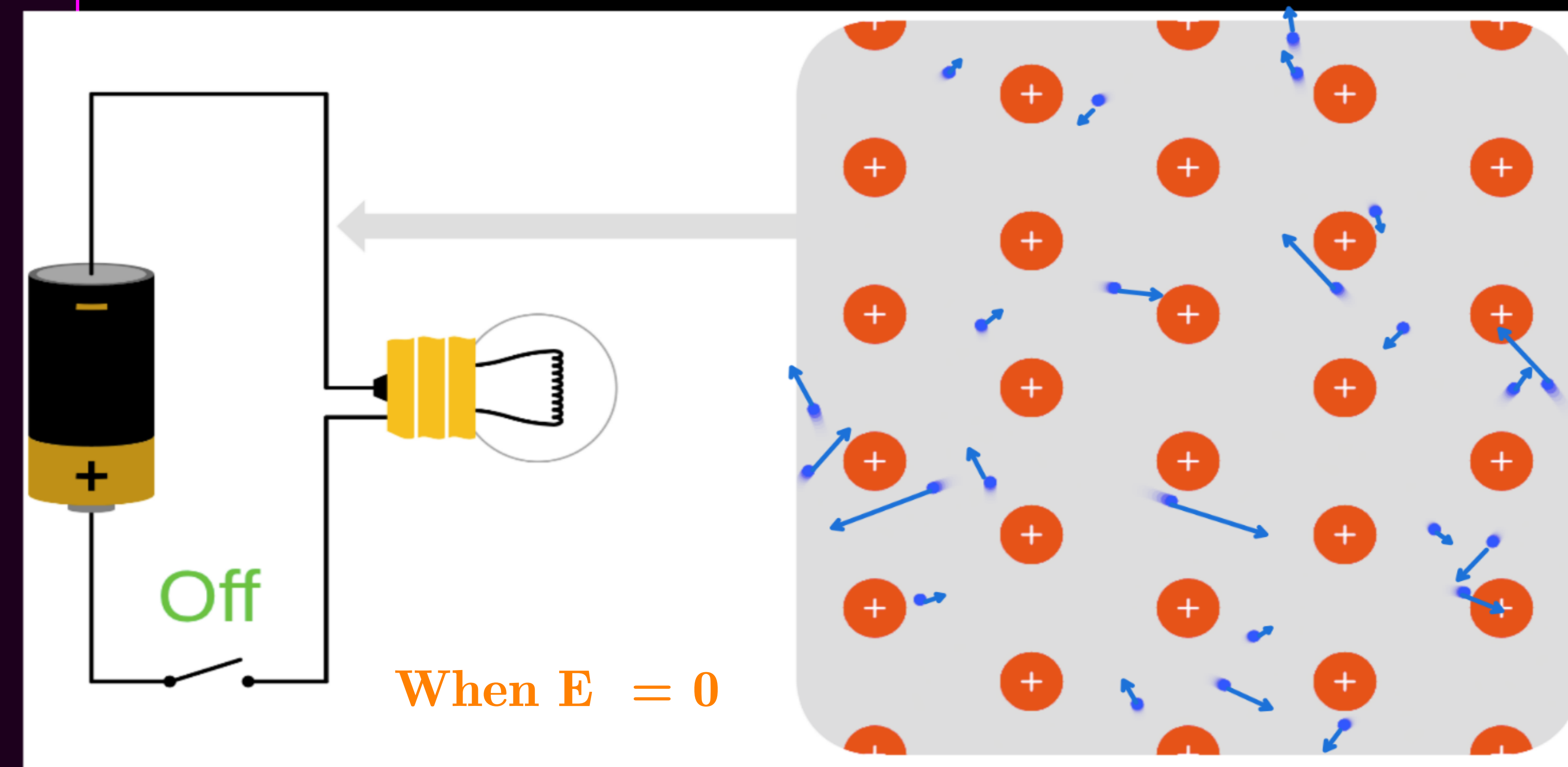
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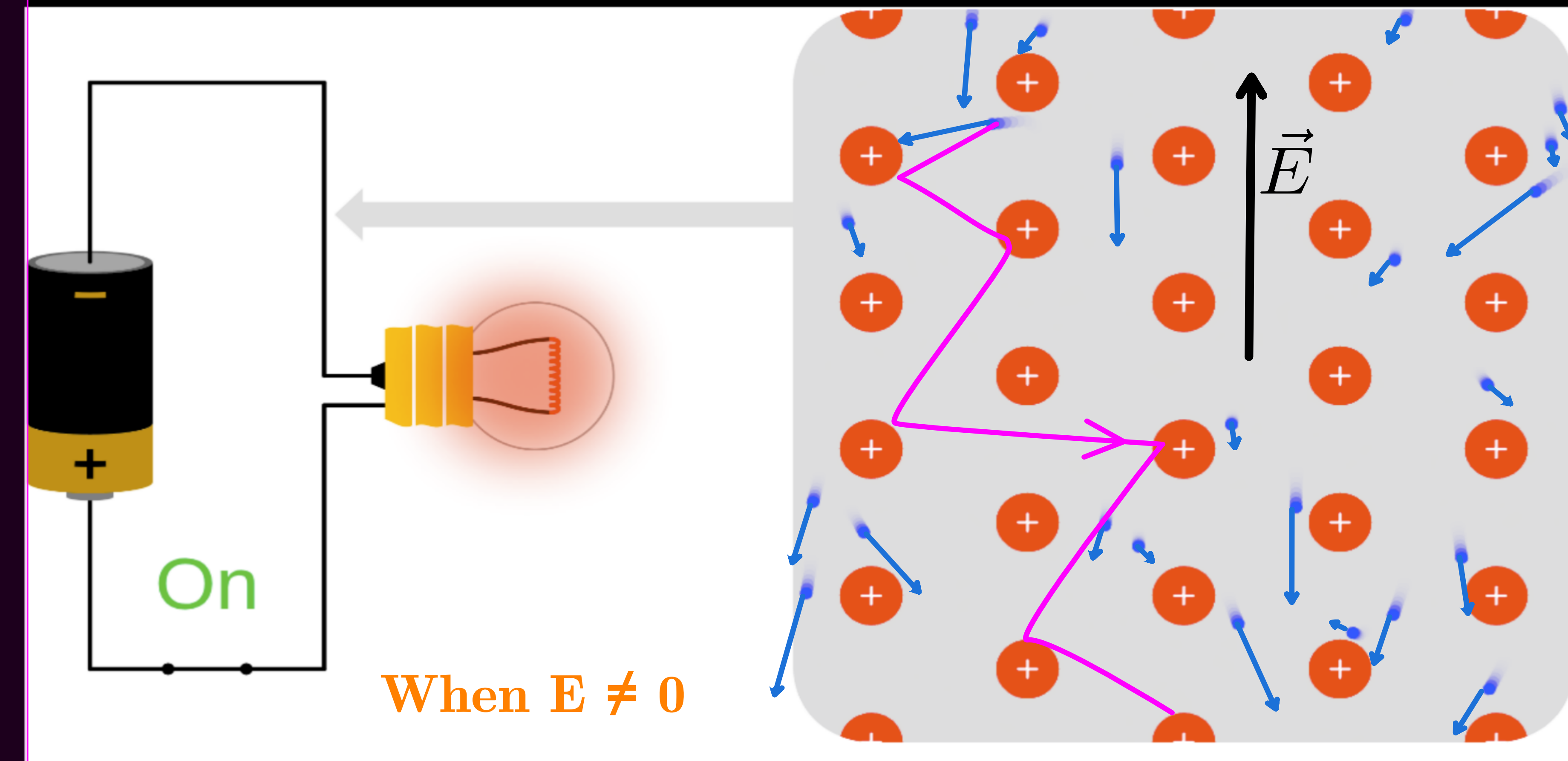
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$$\vec{F} = -e\vec{E} \quad \therefore \text{acceleration } \vec{a} = \frac{-e\vec{E}}{m_e}$$



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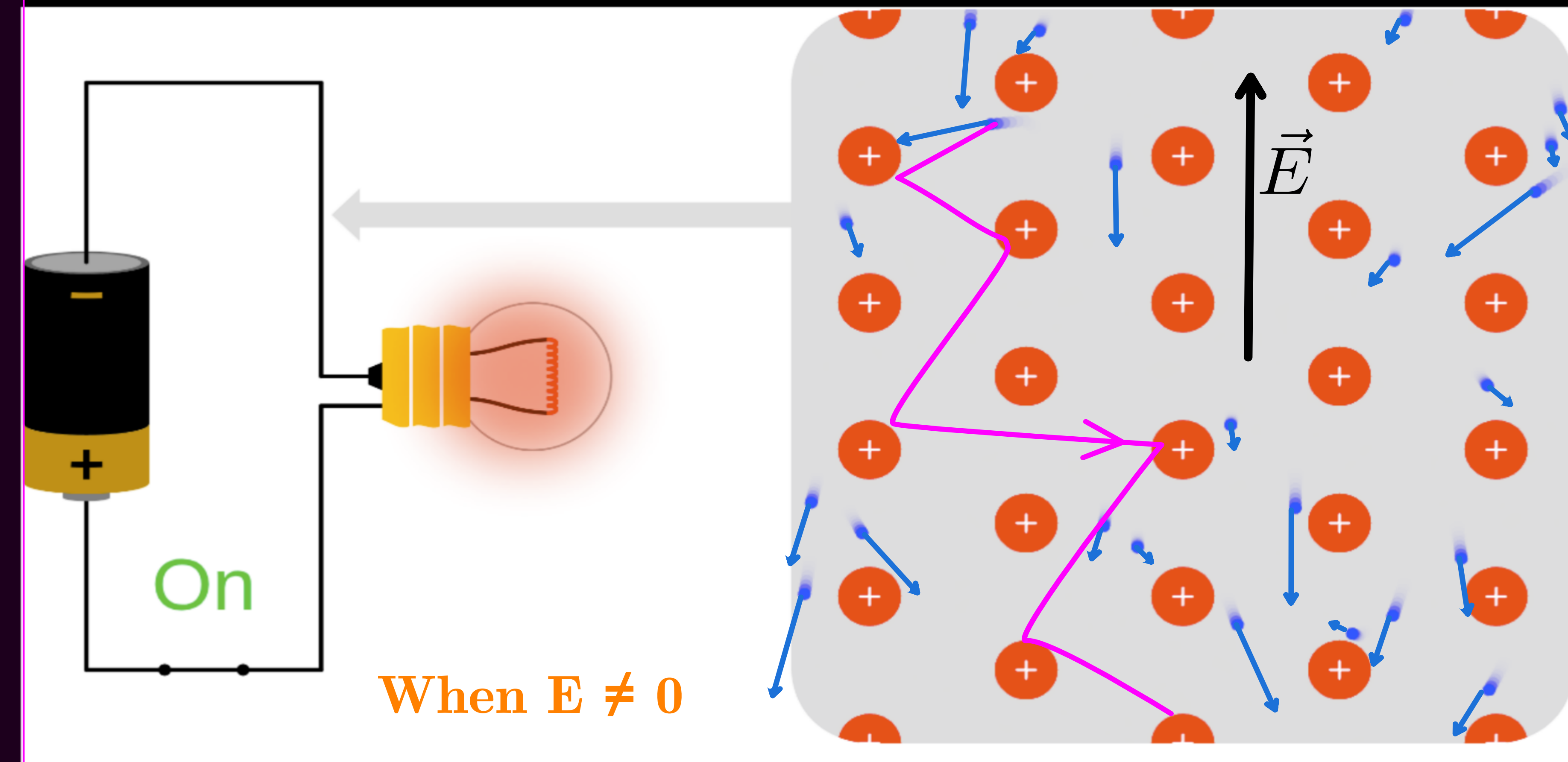
* As the electron accelerates it gains velocity, which last for a short time and is lost in the next collision.



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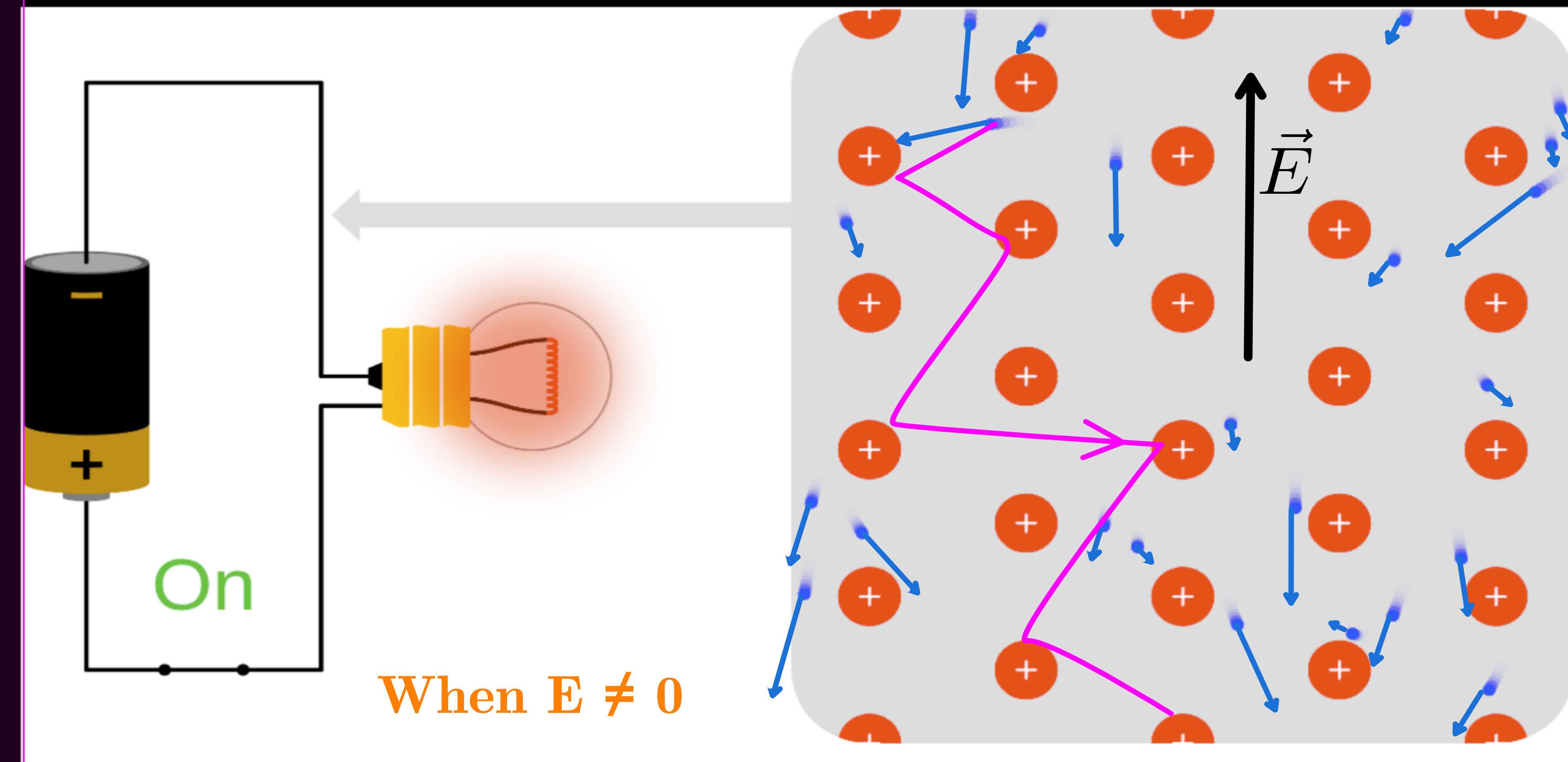


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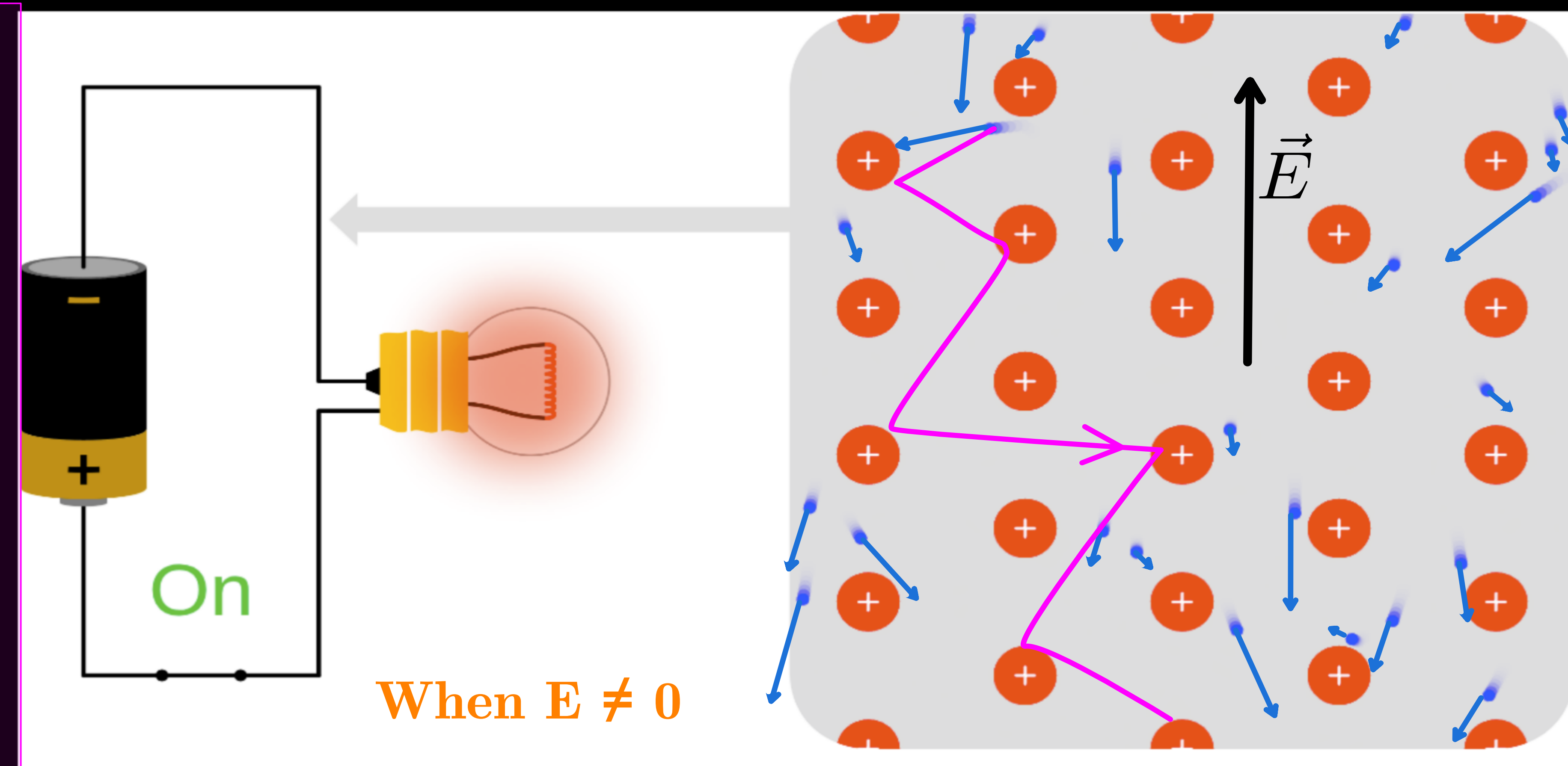


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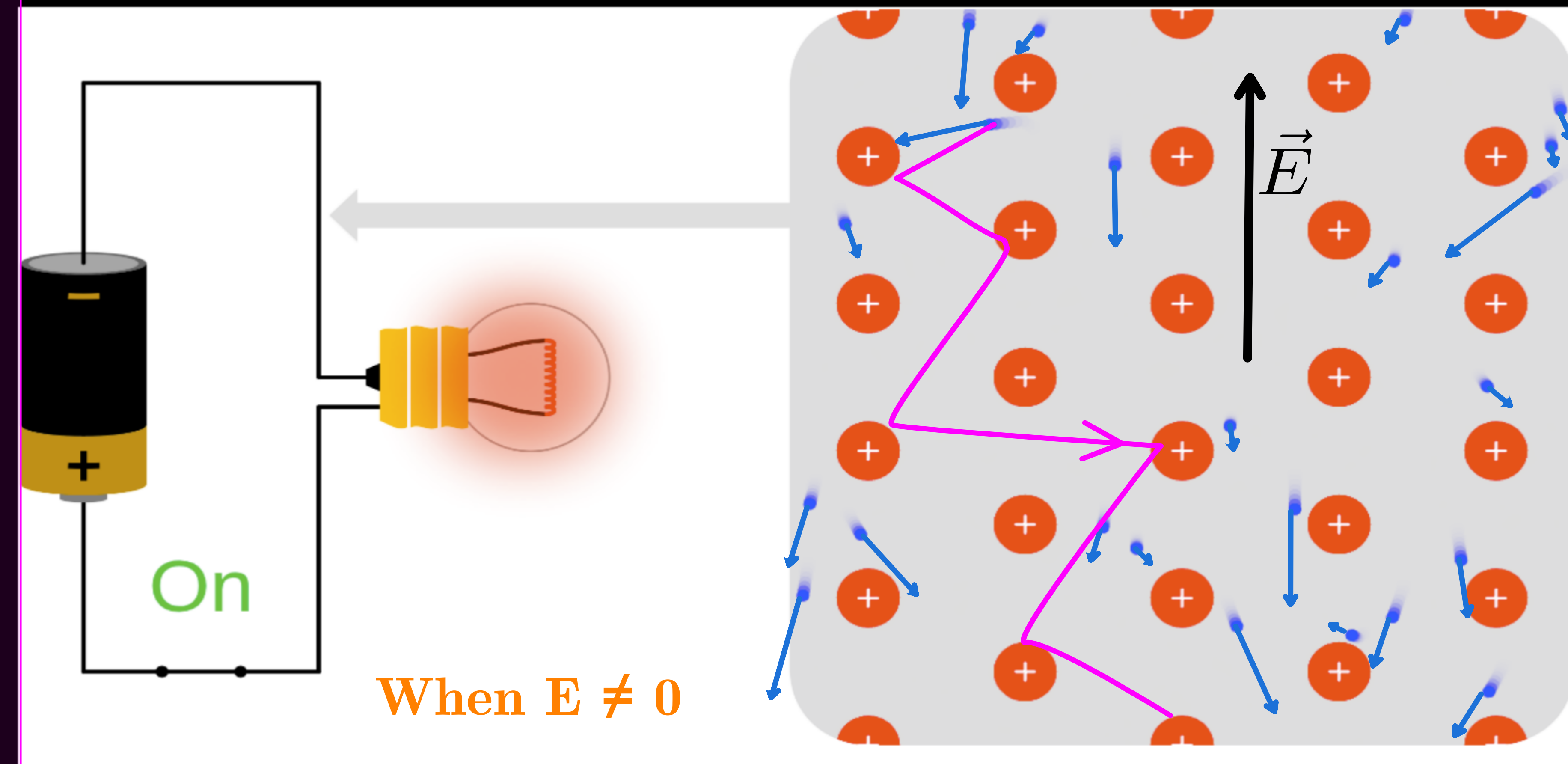
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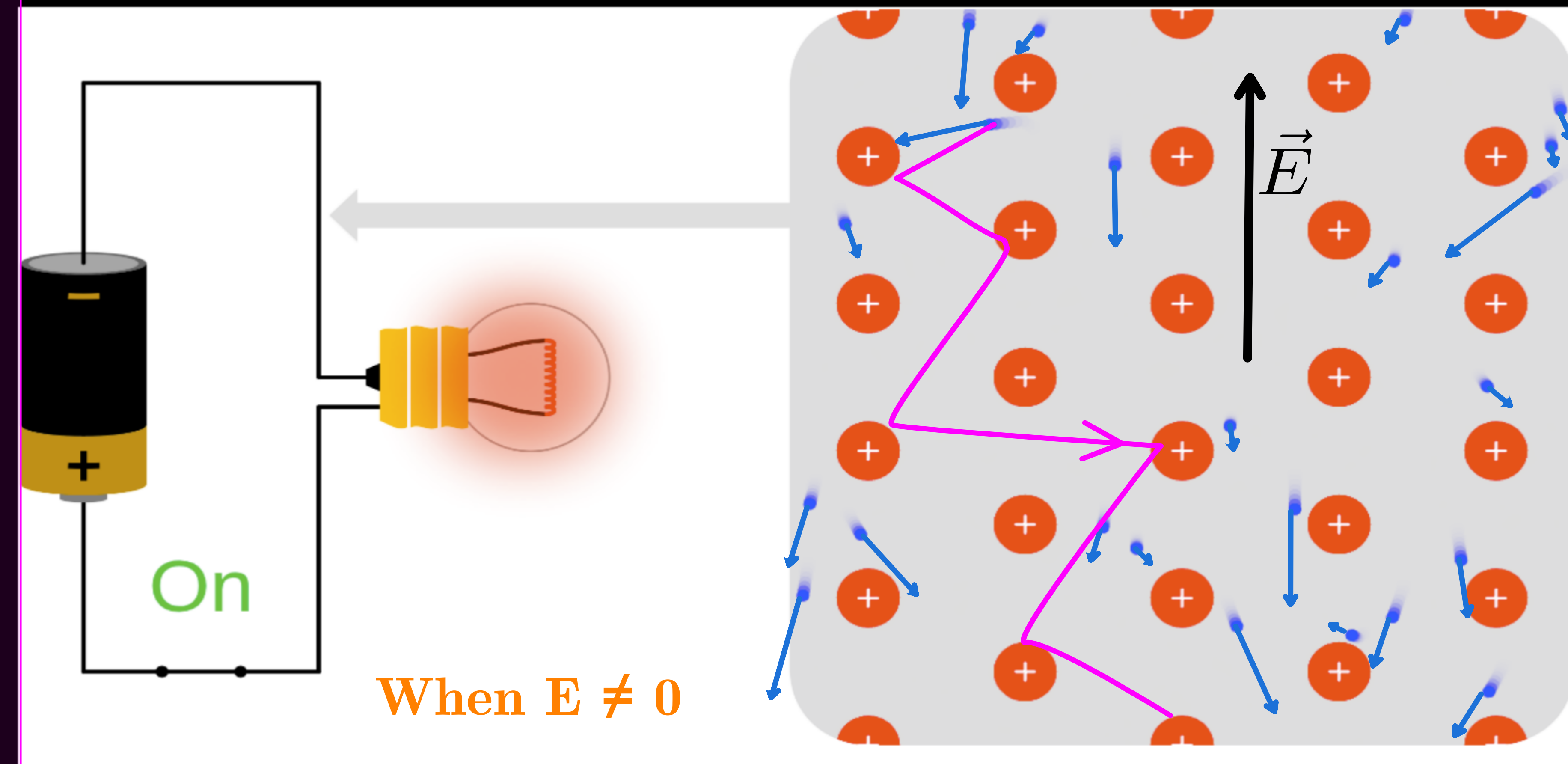
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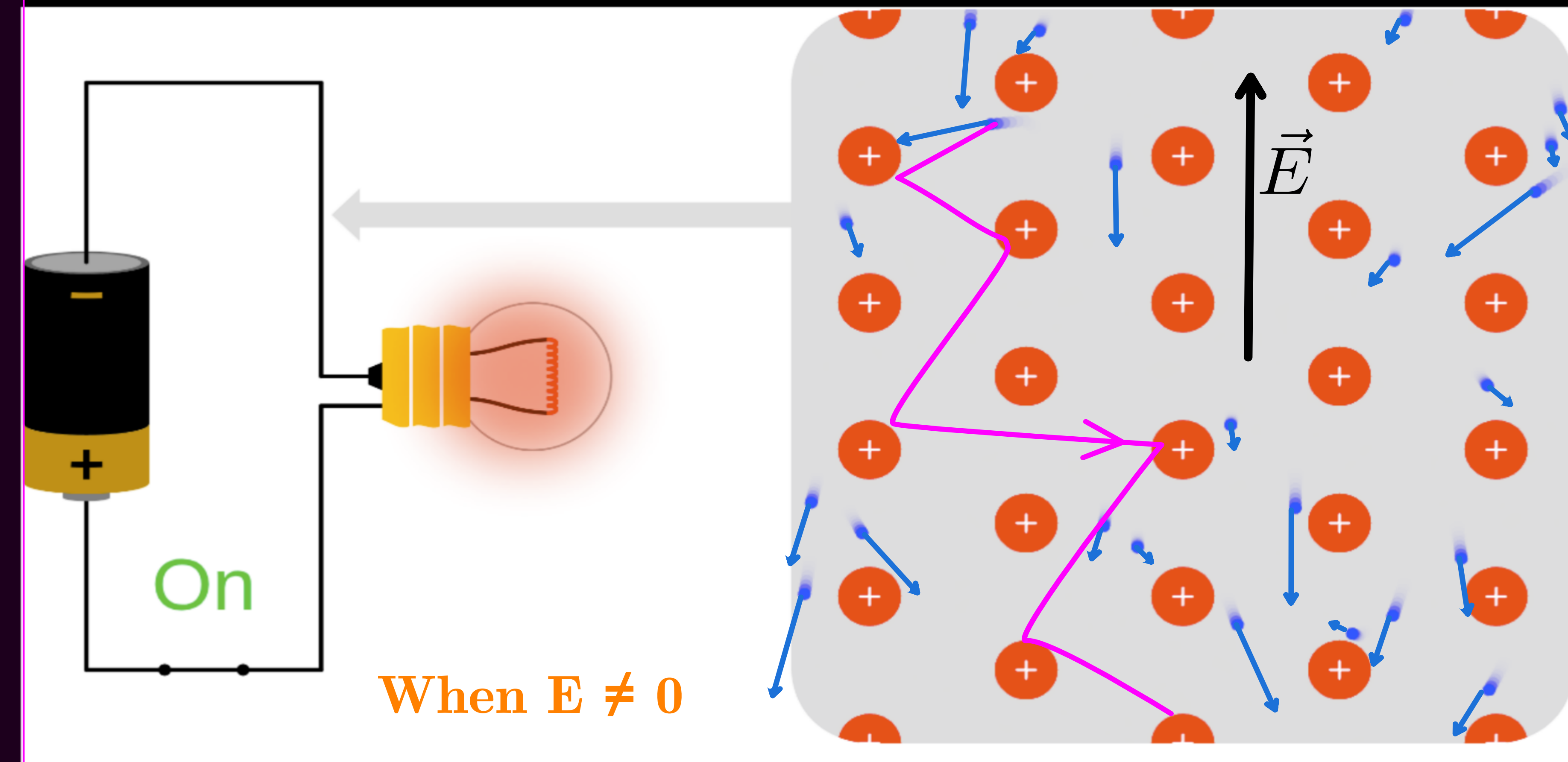
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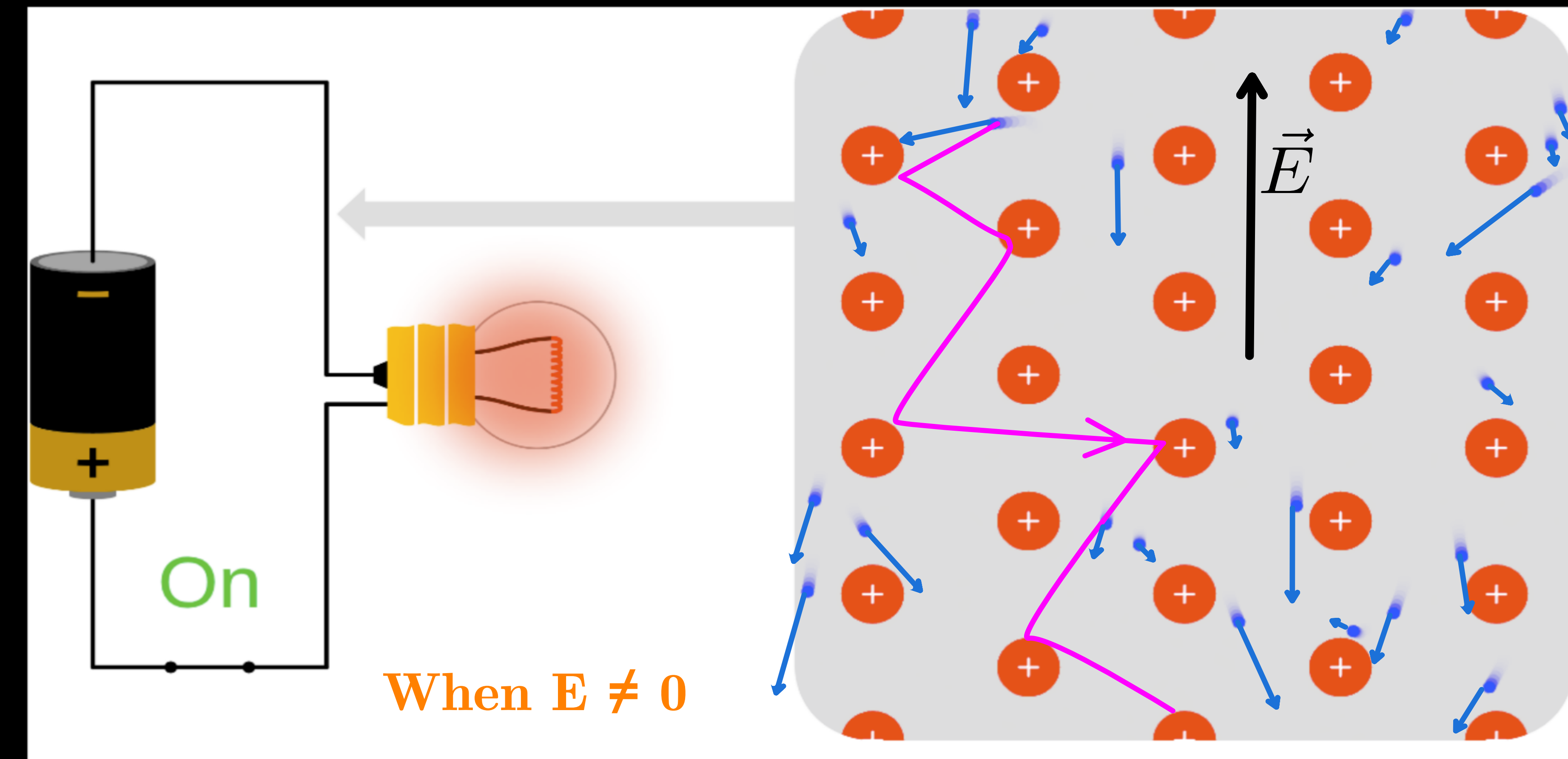
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- * **Drift velocity** : The average velocity with which the free electrons get drifted in the opposite direction of the applied electric field.

DRIFT VELOCITY

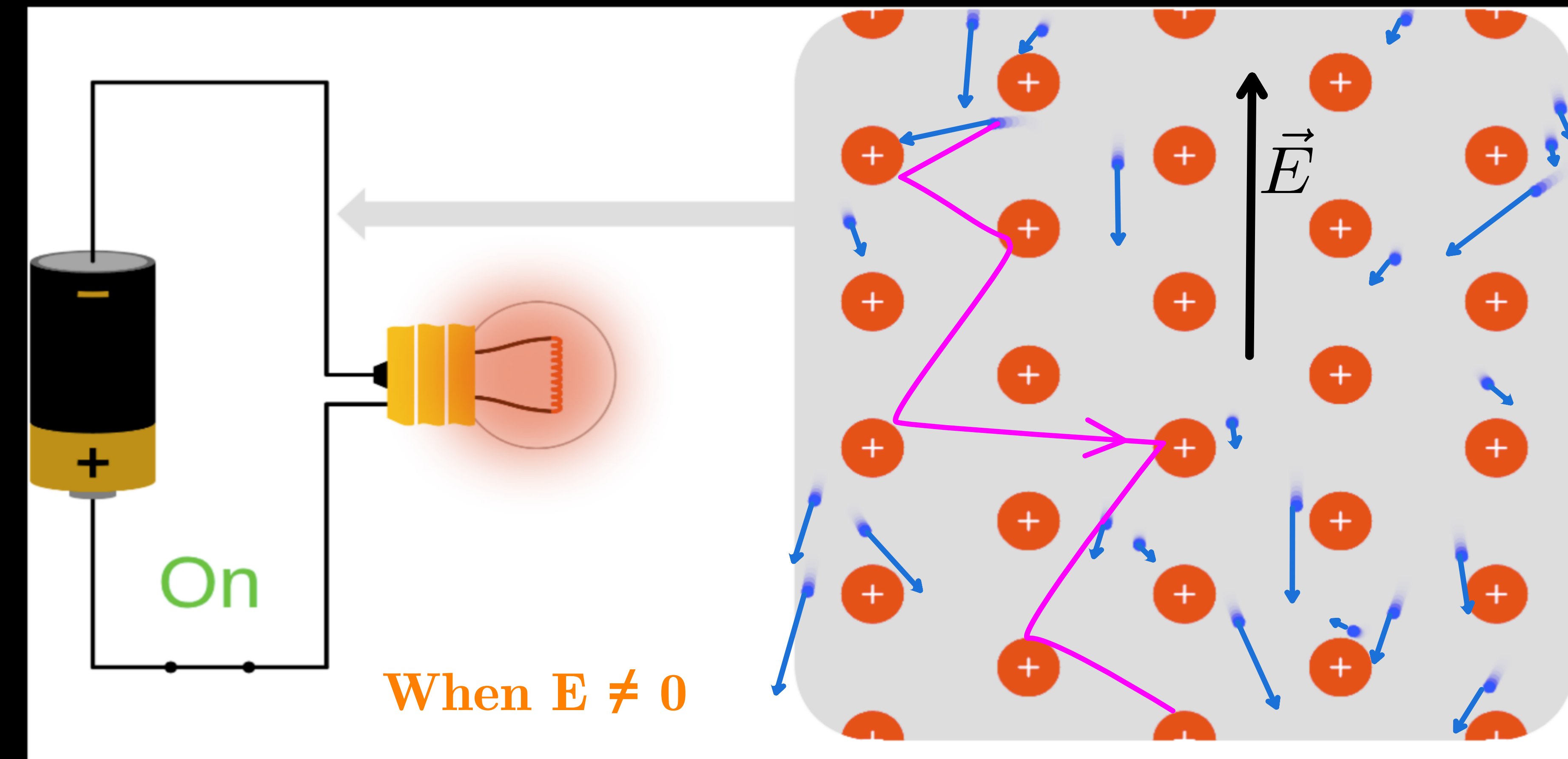
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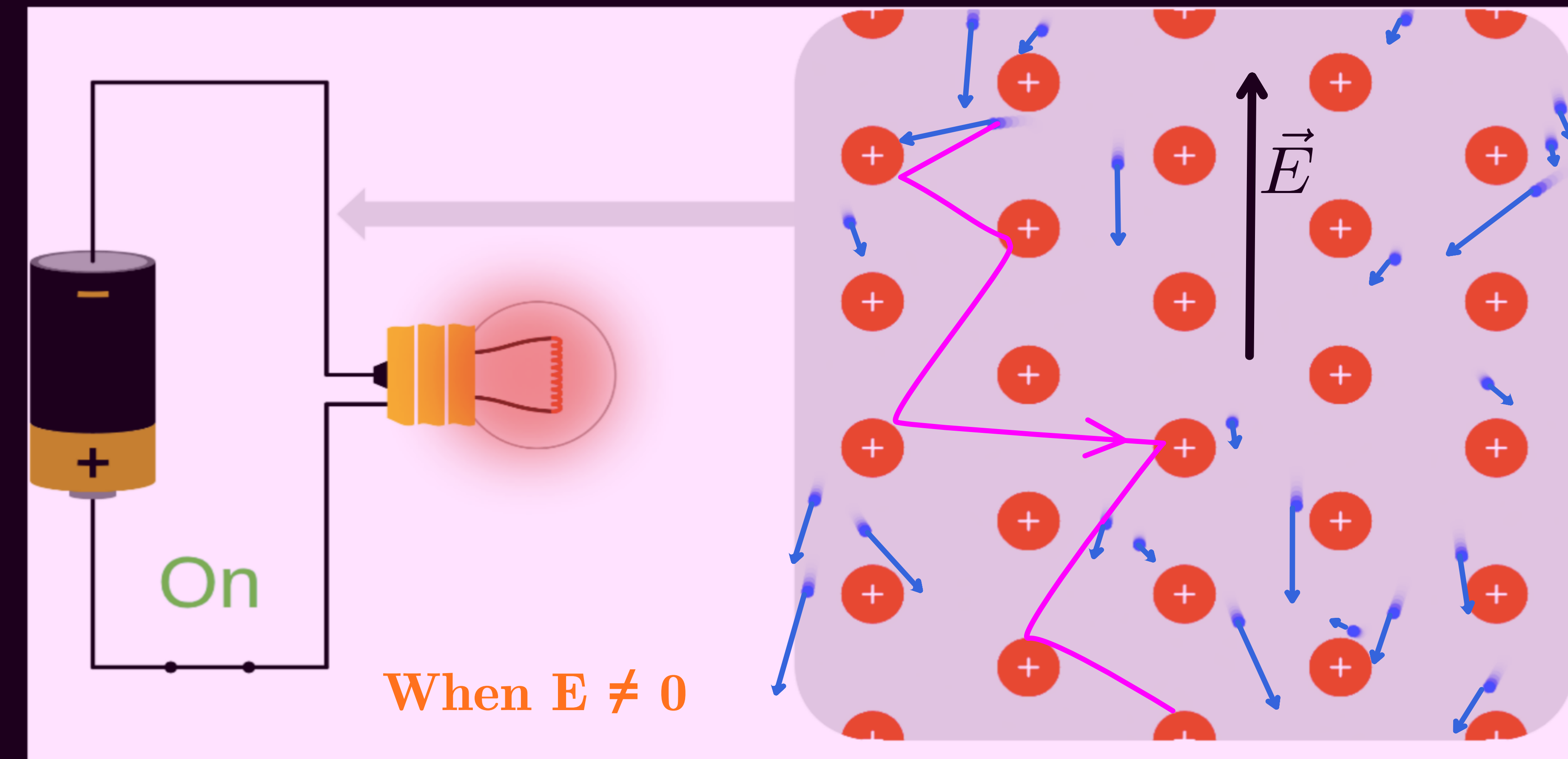


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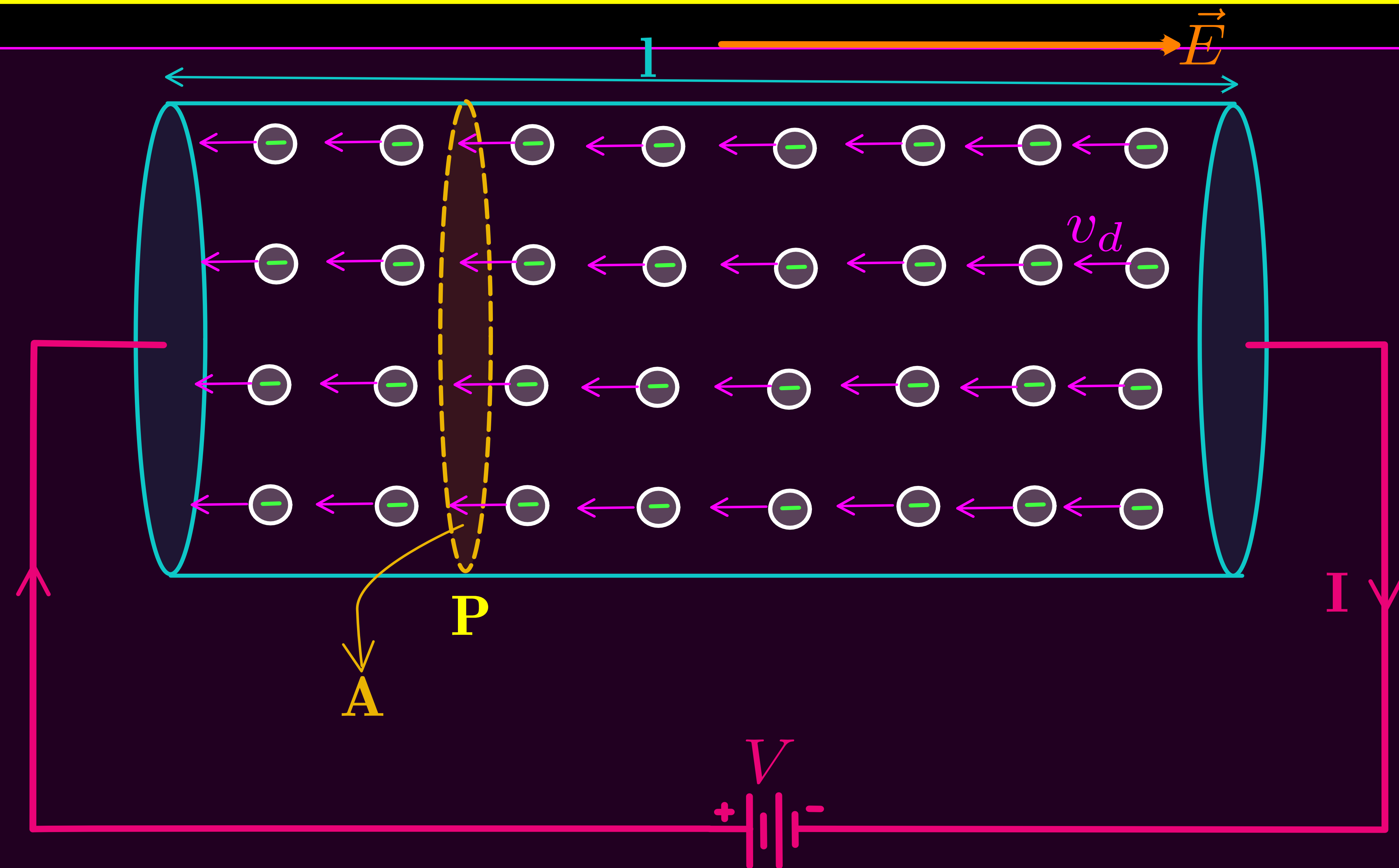
* The drift velocity of electrons is very small in the order of 10^{-4} ms^{-1} .



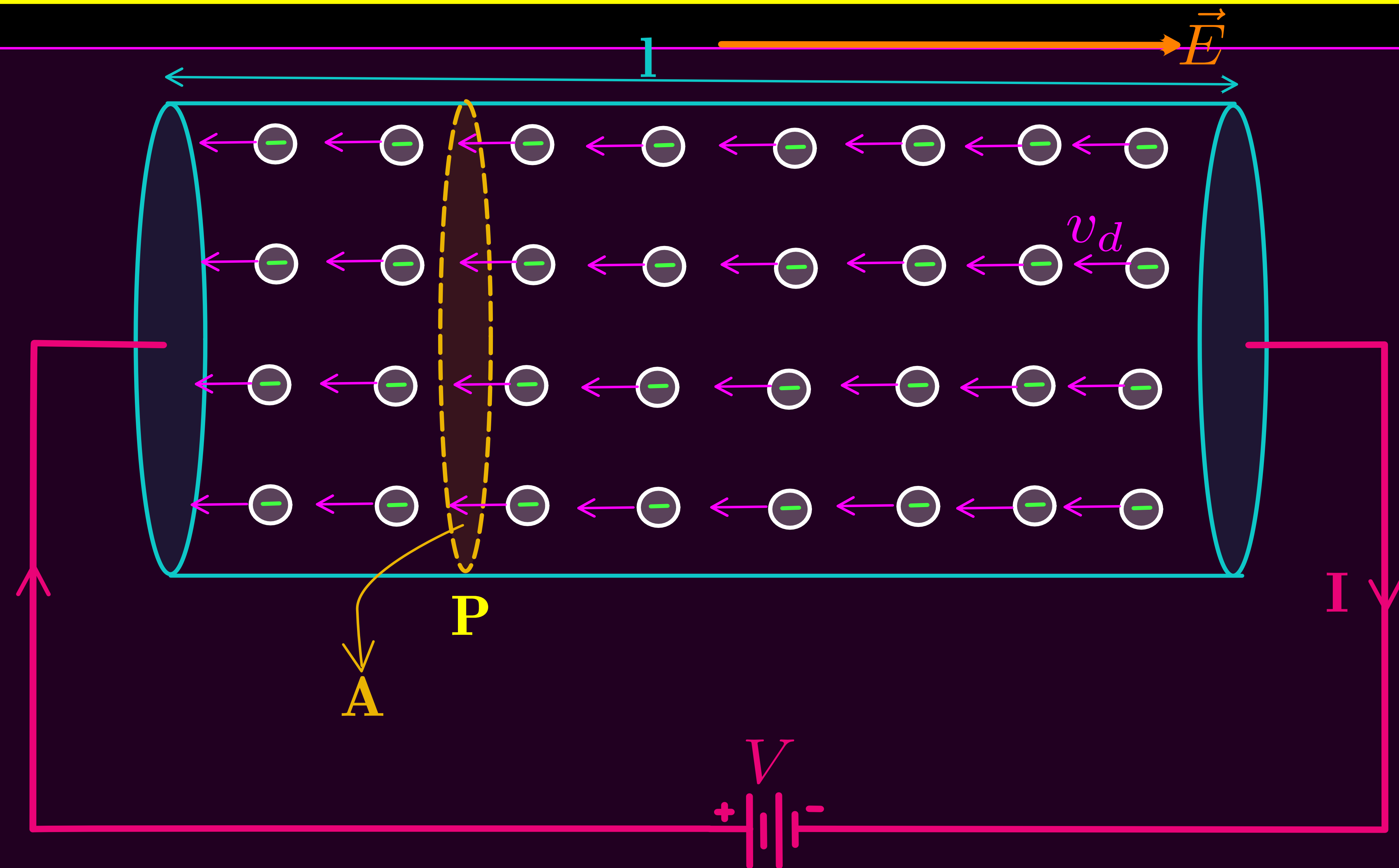
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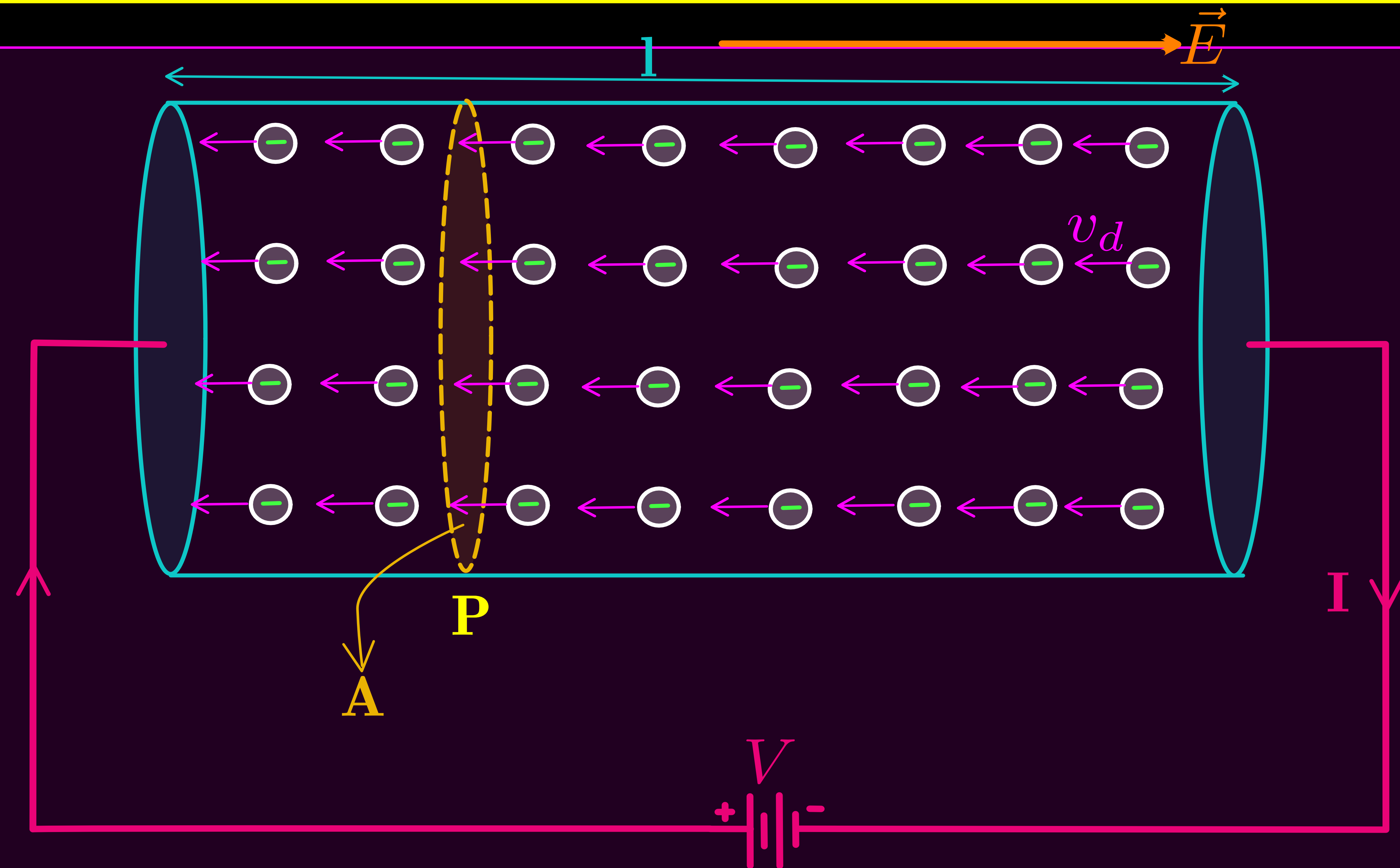


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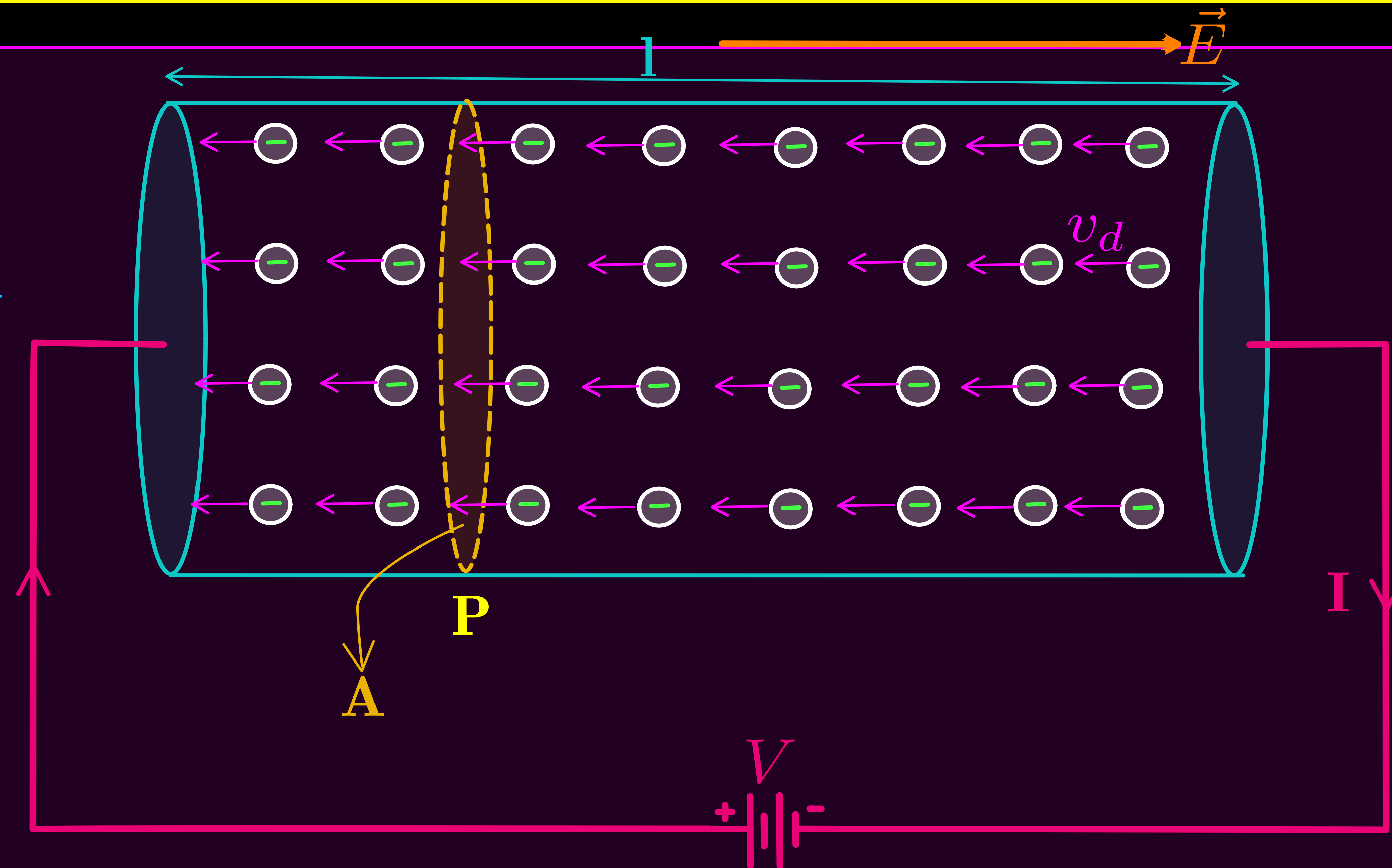
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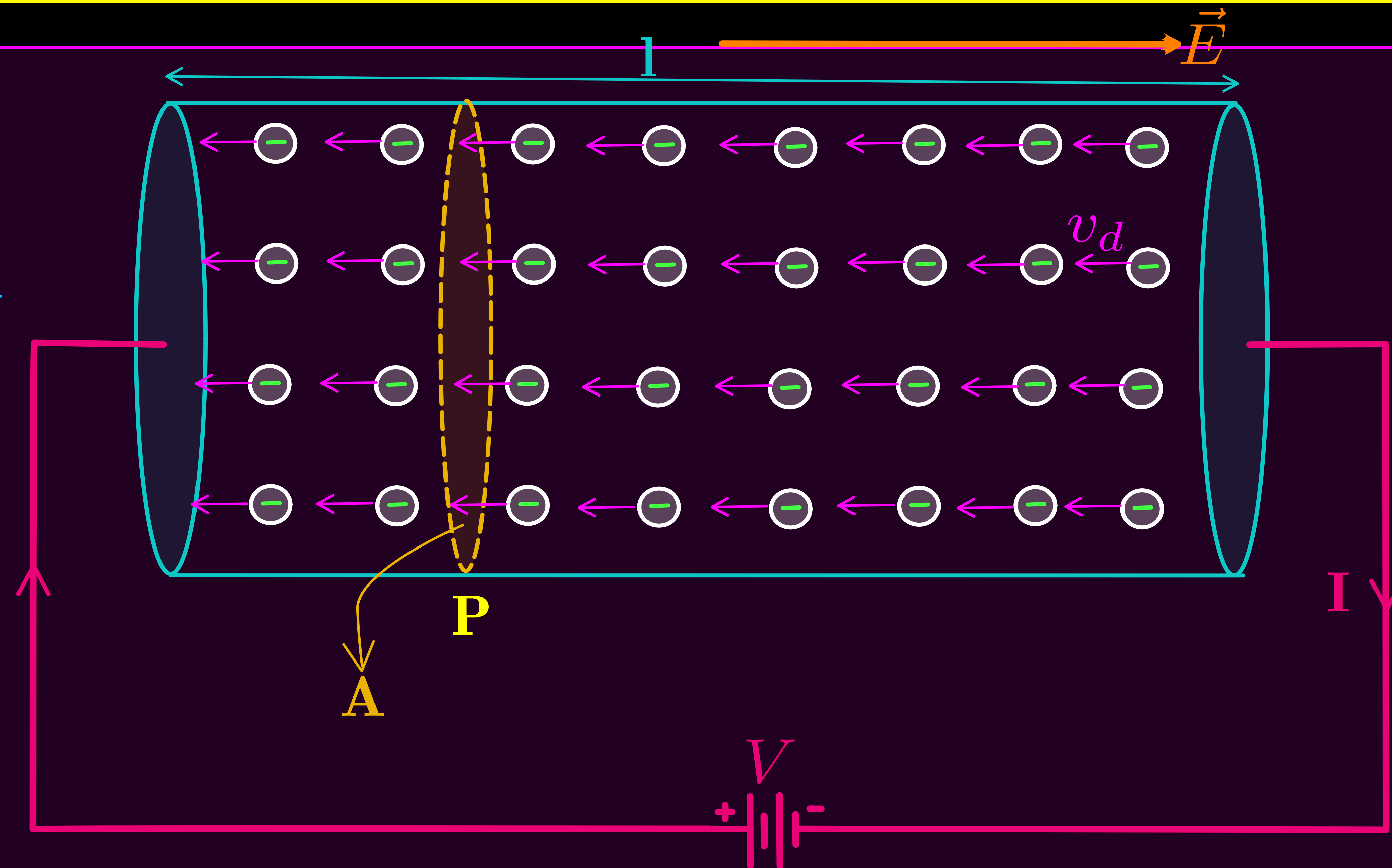
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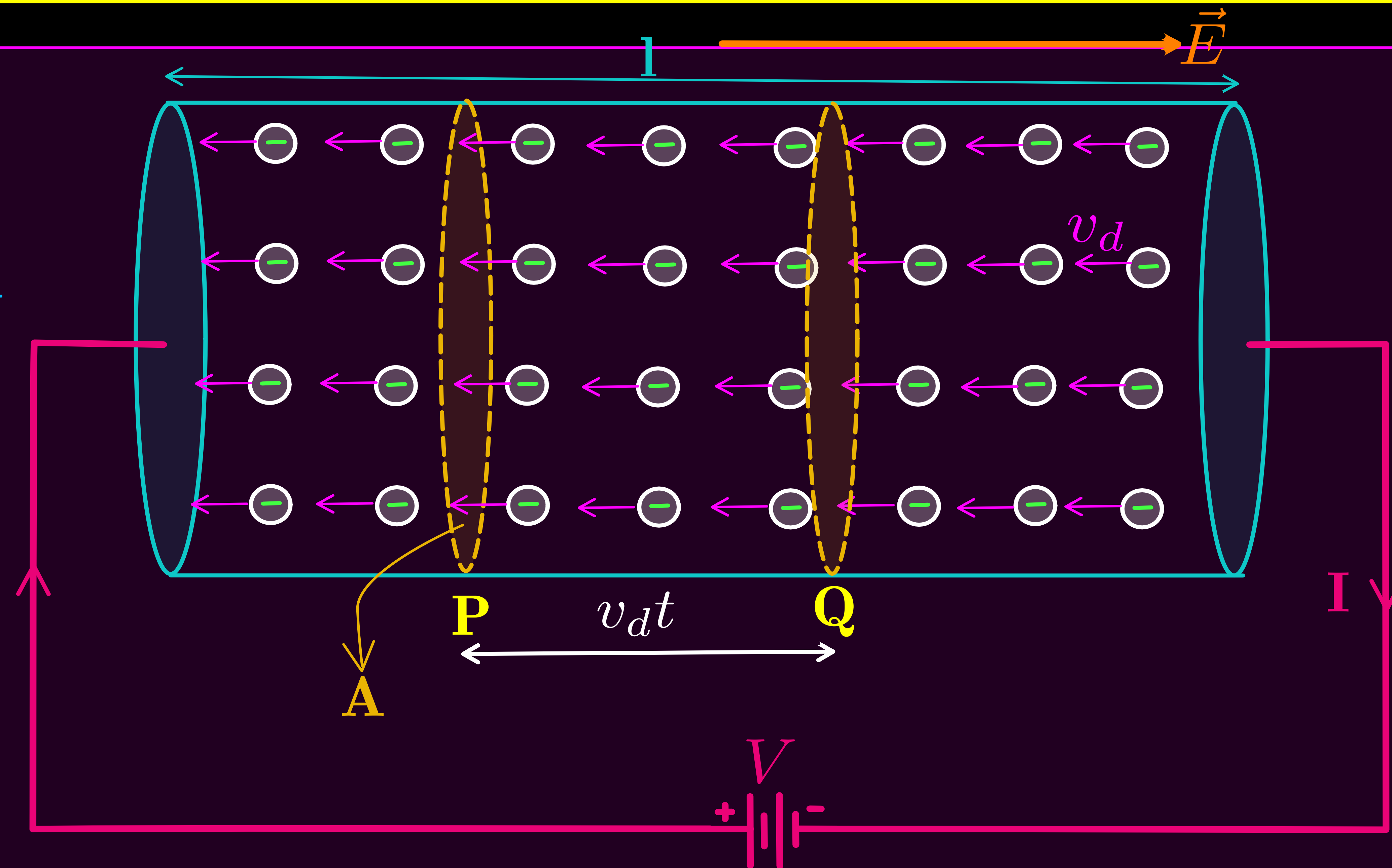
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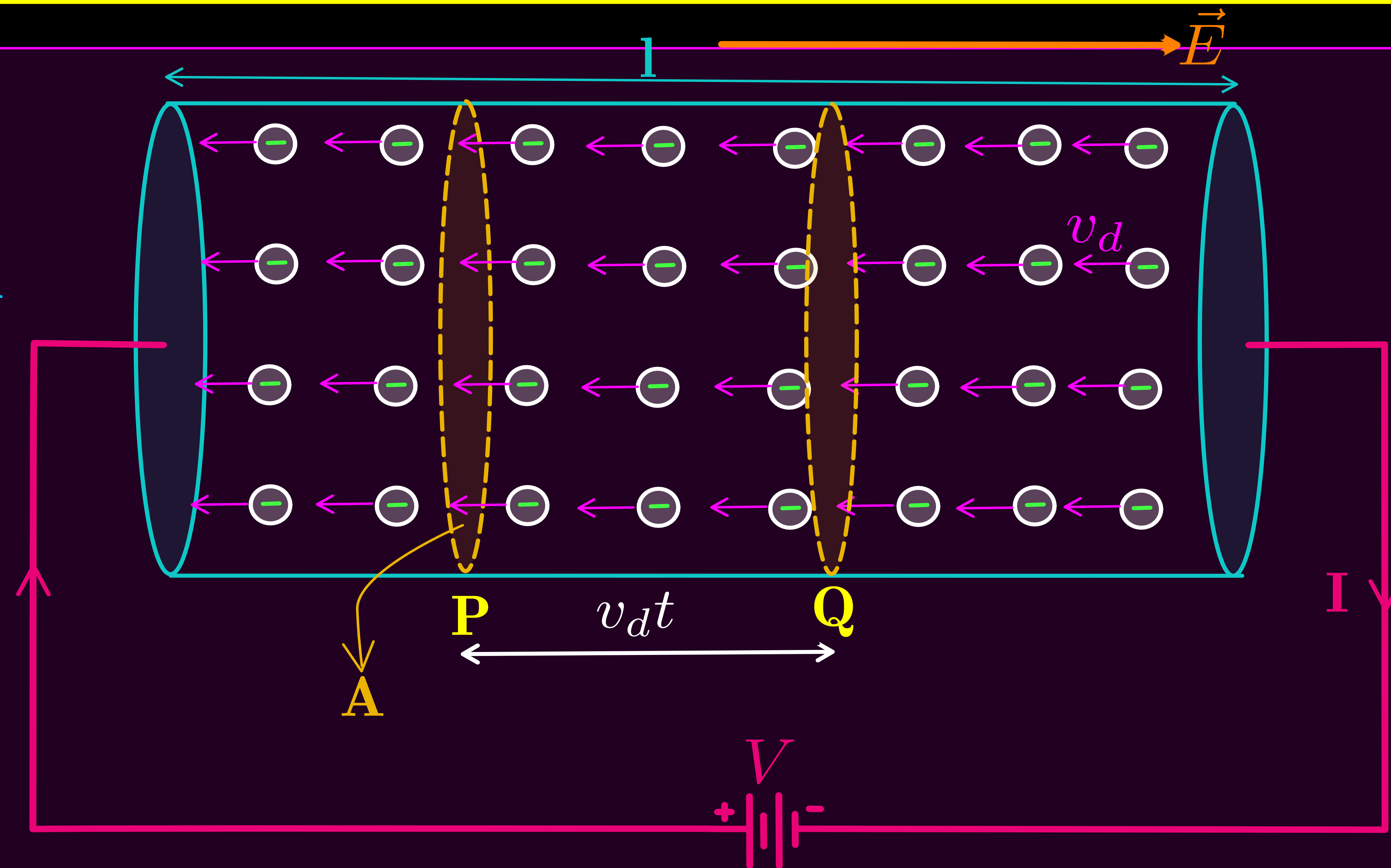
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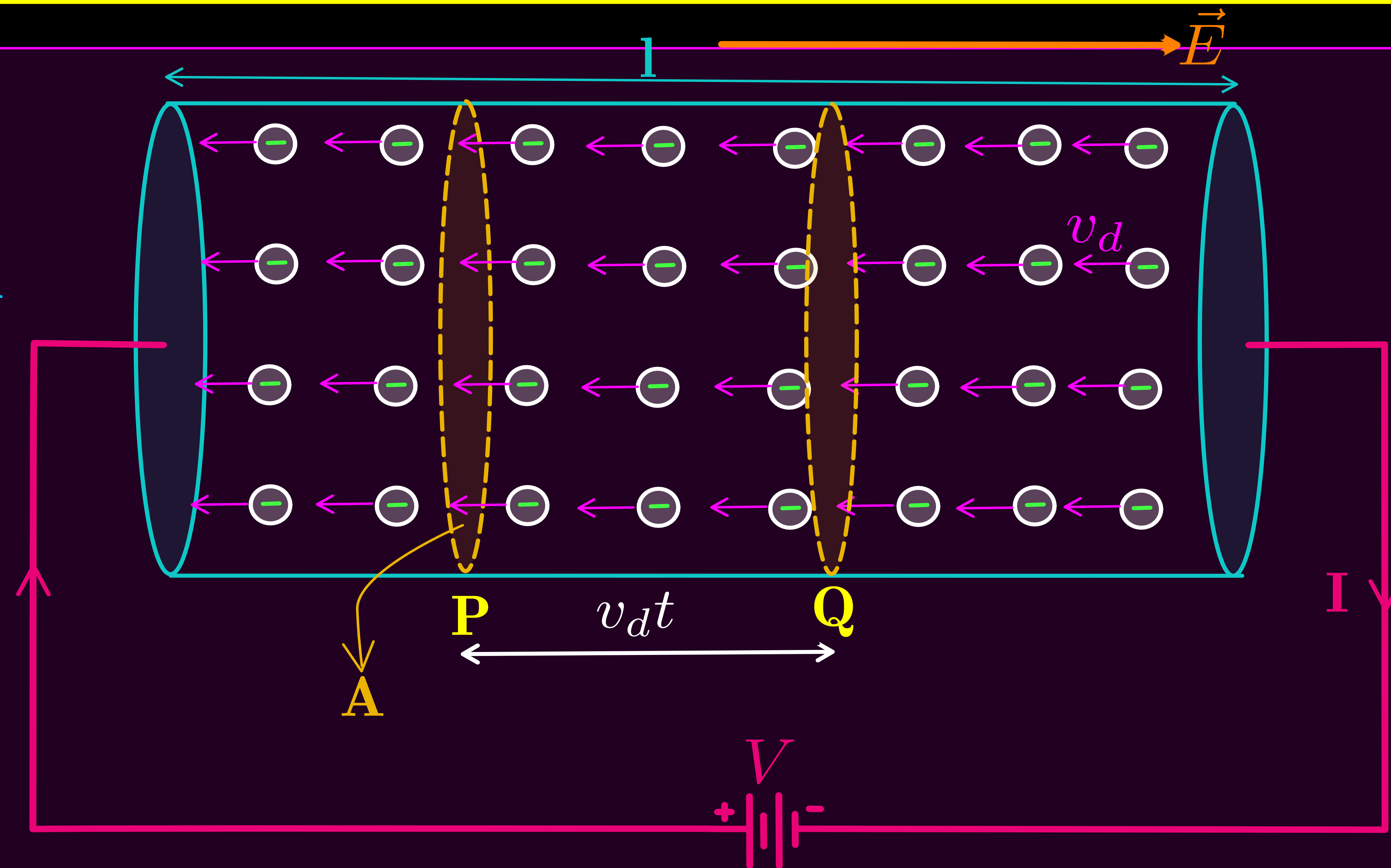
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- * Therefore, number of electron crossing the plane area P in time $t = n \times \text{Volume of cylinder } PQ$

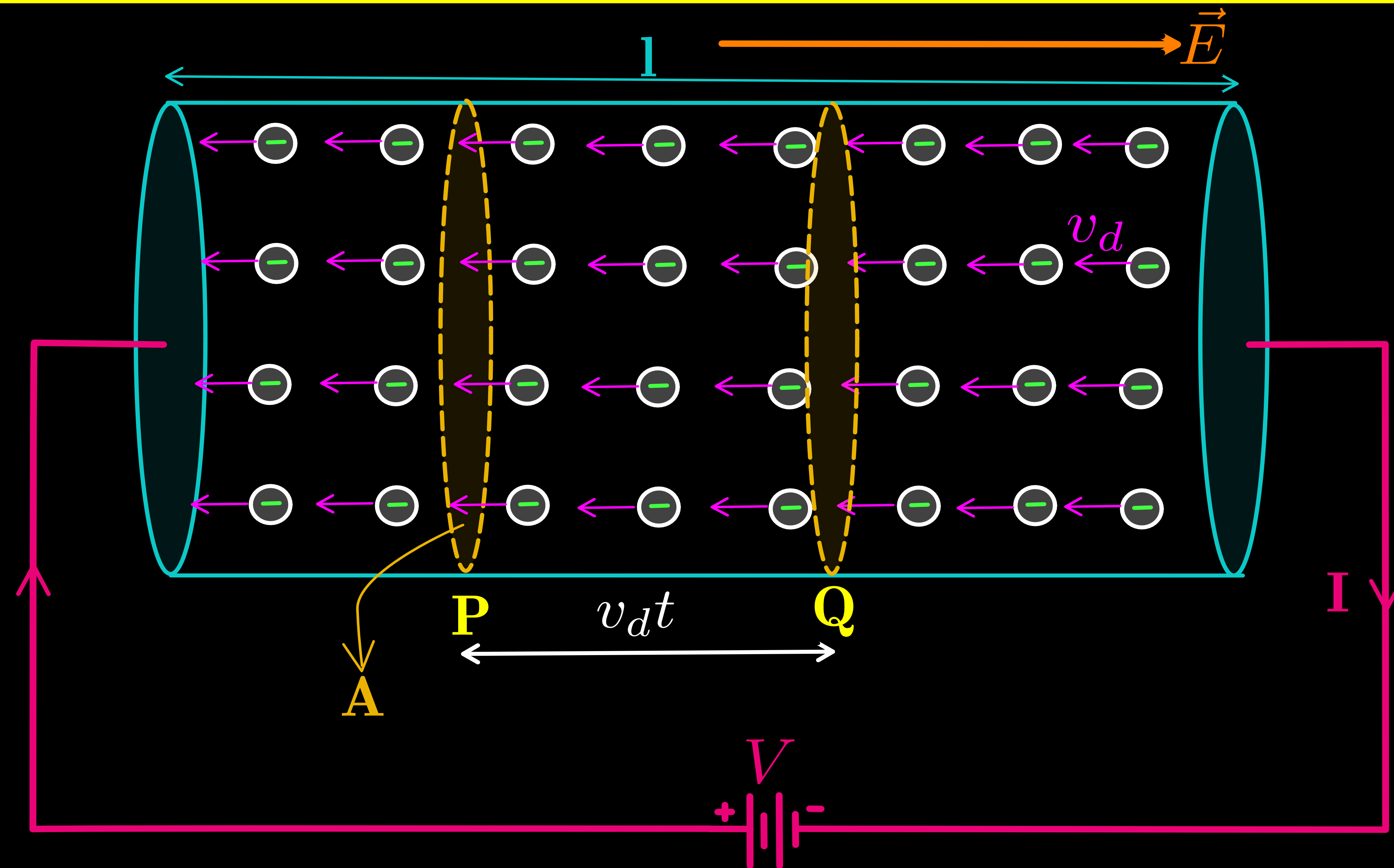
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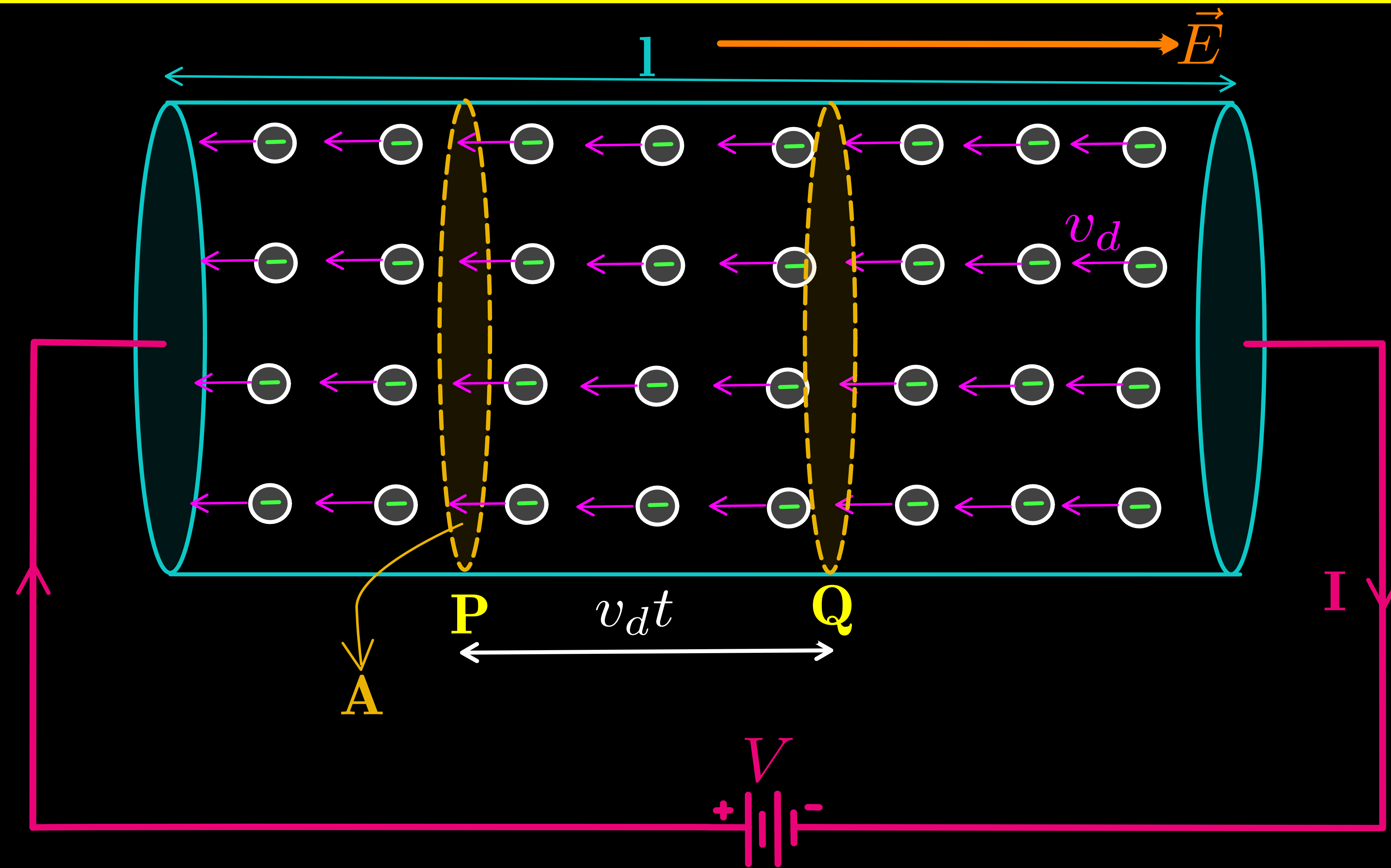
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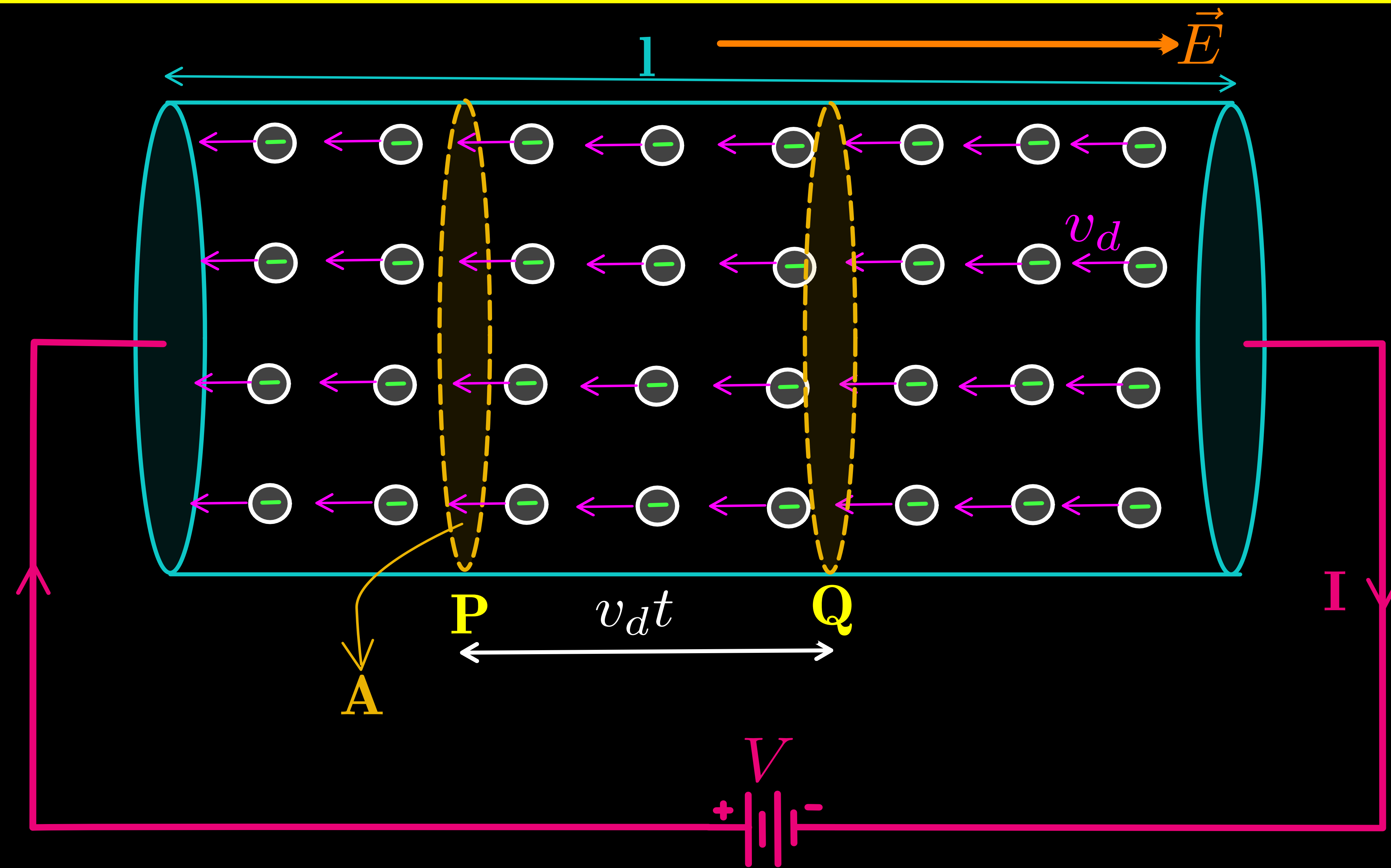
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Ex 9: There is a current of 40 A in a wire of 10^{-6} square metre area of cross-section, if the number of free electrons per cubic metre is 10^{29} , then the drift velocity will be

(a) $1.25 \times 10^3 \text{ m/s}$

(b) $2.5 \times 10^{-3} \text{ m/s}$

(c) $25 \times 10^{-3} \text{ m/s}$

(d) $25 \times 10^3 \text{ m/s}$

Example 3.1 (a) Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area $1.0 \times 10^{-7} \text{ m}^2$ carrying a current of 1.5 A. Assume that each copper atom contributes roughly one conduction electron. The density of copper is $9.0 \times 10^3 \text{ kg/m}^3$, and its atomic mass is 63.5 u. (b) Compare the drift speed obtained above with, (i) thermal speeds of copper atoms at ordinary temperatures, (ii) speed of propagation of electric field along the conductor which causes the drift motion.